

SIEMENS

AK 1703 ACP

Operation and Maintenance

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Note

Please observe Notes and Warnings for your own safety in the Preface.

Disclaimer of Liability

Although we have carefully checked the contents of this publication for conformity with the hardware and software described, we cannot guarantee complete conformity since errors cannot be excluded. The information provided in this manual is checked at regular intervals and any corrections that might become necessary are included in the next releases. Any suggestions for improvement are welcome.

Subject to change without prior notice.

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Preface

This Document Is Applicable To The Following Product(S):

- AK 1703 ACP

Purpose Of This Manual

This manual describes how to parameterize a TM 1703 ACP system, which errors may occur and which are the existing diagnostic options.

Thereby the following points are described step-by-step:

- Safety instructions
- Overview of the Hardware
- Introduction of the engineering tools and their main functions
- Parameter setting of AK 1703 ACP
- Load and save parameters
- Import and export of parameters
- Programming and test
- Load and save of a user program
- Diagnostic
- Display of process states
- Module handling
- Meaning of the LEDs

Target Group

The document you are reading right now is addressed to users, who are in charge of the following engineering tasks:

- Conceptual activities, as for example design and configuration
- Creation of the assembly technical documentation
- System parameterization and system diagnostic
- Utilization of the designated engineering tools
- Technical system maintenance

Within this manual there are hints how to obtain information or files by means of *PRISMA* (<http://intranet.isi-ptdea.siemens.com>). Should *PRISMA* not be available to you please consult your contact person at Siemens.

Conventions

- Manuals to be referred to are represented in italics, such as e.g. Common Functions System and Basic System Elements, section "Information Objects".
- Inputs via keyboard or mouse keys and visible on the screen are described in this font.
- Menu positions visible on the screen ARE DESCRIBED IN THIS FONT.
- Operation code of programs and variable names .
- Symbolic names, program names and file name extensions *are represented in this font.*



Note

The functions for operation, diagnostic and test of the TM 1703 ACP described in this manual have been documented based on a PC with the Windows 2000 Professional operating system. They are also applicable - under some circumstances with inessential deviations - for other operating

Recommendations For Third-Party Products

Siemens does neither receive liability nor warranty for recommendations which are given or implied by this manual. For the correct and intended use of the respective product the associated technical descriptions must be paid attention to in any case.

Links To Third-Party Websites

Siemens is not responsible for the contents of third-party websites mentioned in this document, as well as the correctness of the publications and links. For all product information the respective manufacturer is responsible.

Notes On Safety

This manual does not constitute a complete catalog of all safety measures required for operating the equipment (module, device) in question because special operating conditions might require additional measures. However, it does contain notes that must be adhered to for your own personal safety and to avoid damage to property. These notes are highlighted with a warning triangle and different keywords indicating different degrees of danger.



Danger

means that death, serious bodily injury or considerable property damage **will** occur, if the appropriate precautionary measures are not carried out.



Warning

means that death, serious bodily injury or considerable property damage **can** occur, if the appropriate precautionary measures are not carried out.

Caution

means that minor bodily injury or property damage could occur, if the appropriate precautionary measures are not carried out.



Note

is important information about the product, the handling of the product or the respective part of the documentation, to which special attention is to be given.



Qualified Personnel

Commissioning and operation of the equipment (module, device) described in this manual must be performed by qualified personnel only. As used in the safety notes contained in this manual, qualified personnel are those persons who are authorized to commission, release, ground, and tag devices, systems, and electrical circuits in accordance with safety standards.

Use as Prescribed

The equipment (device, module) must not be used for any other purposes than those described in the Catalog and the Technical Description. If it is used together with third-party devices and components, these must be recommended or approved by Siemens.

Correct and safe operation of the product requires adequate transportation, storage, installation, and mounting as well as appropriate use and maintenance.

During operation of electrical equipment, it is unavoidable that certain parts of this equipment will carry dangerous voltages. Severe injury or damage to property can occur if the appropriate measures are not taken:

- Before making any connections at all, ground the equipment at the PE terminal.
- Hazardous voltages can be present on all switching components connected to the power supply.
- Even after the supply voltage has been disconnected, hazardous voltages can still be present in the equipment (capacitor storage).
- Equipment with current transformer circuits must not be operated while open.
- The limit values indicated in the manual or the operating instructions must not be exceeded; that also applies to testing and commissioning.

Consider obligatory the safety rules for the accomplishment of works at electrical plants:

1. Switch off electricity all-pole and on all sides!
2. Ensure that electricity cannot be switched on again!
3. Double check that no electrical current is flowing!
4. Discharge, ground, short circuit!
5. Cover or otherwise isolate components that are still electrically active!

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1. Overview

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1.1. General

AK 1703 ACP is an automation and telecontrol system with a standardized communication interface. It provides parameter-settable functions for the preprocessing and postprocessing of input and output signals, configurable communication protocols, and also a programmable logic controller for the realization of control-technical tasks.

All functions provided by AK 1703 ACP can be configured with the help of the TOOLBOX II:

- Project technique
 - Customer
 - Plant
- System technique
 - Automation unit
 - HW configuration
 - Communication
- Assembly technique
- Process technique
 - Periphery settings
 - Preprocessing of signals
 - Postprocessing of signals
 - Open-/closed-loop control function (function diagram)

With the TOOLBOX II, operating functions which serve for commissioning and service can also be carried out in the running automation system (online):

- Loading of system parameters
- Loading of telecontrol parameters
- Loading of application programs
- Test
 - Data flow test
 - Message simulation
 - Online test
 - Topology test
- Diagnostic

1.2. System Architecture

1.2.1. System Components

AK 1703 ACP forms an automation unit of the system family SICAM 1703 and is structured from the following components:

- Mounting rack
- Power supply modules
- Basic system elements
 - Master control element including flash card (necessary)
 - Processing and communication elements (optional)
- Additional system elements
 - Configurable protocol elements for the communication with other automation units (configurable on a basic system element)
 - Peripheral elements
 - locally in the mounting rack
 - remotely via the Ax 1703 peripheral bus
- Connection boards for connecting the communication interfaces
- Bus interfaces for connecting remote peripheral elements

Power Supply

Typ	Bezeichnung	Power
PS-5620	Power Supply 24-60 VDC	80 W
PS-5622	Power Supply 110-220 VDC, 230 VAC	80 W

Depending on the chosen configuration, 1 up to 4 power supply modules can be configured. Exact details thereto can be found in the manual *AK 1703 ACP Installation, chapter "Power Supply"*.

Power Supply Peripheral Elements TM 1703 ACP

Type	Designation	Power
PS-6630	Power Supply Module 24-60 VDC (EMC+)	8.0 W
PS-6632	Power Supply Module 110-220 VDC (EMC+)	8.0 W

Configuration Master Control Element

Type	Designation	
CP-2010/CPC25	System Functions, Processing and Communication	required
CM-0843, CM-0842	Bus Interfaces Ax-PE	optional
SM-2545, SM-2551, SM-2556, SM-2556 with SM-0551	Serial Interface Modules (SIM)	optional

Configuration Processing And Communication Element

Type	Designation	
CP-2010/CPC25	System Functions, Processing and Communication	required
CP-2017/PCCX25	Processing & Communication	optional
CM-0843, CM-0842	Bus Interfaces Ax-PE	optional
SM-2545, SM-2551, SM-2556, SM-2556 with SM-0551	Serial Interface Modules (SIM)	optional



Note

Except SM-2545, all SIM can be equipped on each slot (SIM0 and SIM1) of CP 2017 and combined arbitrarily. SM-2545 may only be equipped on card location SIM1, but can arbitrary be combined with every other SIM.

Protocol Elements

Type	Designation	
SM-x551/BPPA0	Standard protocol for point-to-point traffic	optional
SM-x551/UMPMA0	Standard protocol for multi-point traffic (M)	optional
SM-x551/UMPSA0	Standard protocol for multi-point traffic (S)	optional
SM-x551/SFBMA1	Standard protocol for field bus (M)	optional
SM-x551/SFBSA1	Standard protocol for field bus (S)	optional
SM-x551/DIAMA0	Standard protocol for dial-up traffic (M)	optional
SM-x551/DIASA0	Standard protocol for dial-up traffic (S)	optional
SM-x551/103MA0	Standard protocol for interfacing of protective devices (M)	optional
SM-2545/DPM00	Standard protocol for Profibus DP	optional
SM-2556/ET02	Standard protocol for Ethernet TCP/IP IEC 104	optional
SM-2556/ETA2	Standard protocol for Ethernet TCP/IP IEC 104	optional
SM-2556/ET03	Standard protocol for Ethernet TCP/IP IEC 61850	optional

(M) ... Master, (S) ... Slave

Connection Of The Peripheral Elements

Type	Designation	Connection	Number of peripheral elements ^{*)}
---	Local Ax 1703 Peripheral Bus	electrical	CM-2832: up to 7 CM-2835: up to 15
CM-0843	Bus Interface Ax-PE 4x USB	electrical	up to 16
CM-0842	Bus Interface Ax-PE 4x optical	optical	up to 16

^{*)} all together up to 16 peripheral elements per processing and communication element

Peripheral Elements AK 1703 ACP

Type	Designation	
DI-2100/BISI25	Binary signal input (8x8, 24-60VDC)	optional
DI-2110/BISI26	Binary signal input (8x8, 24-60VDC)	optional
DI-2111/BISI26	Binary signal input (8x8, 110/220VDC)	optional
DO-2201/BISO25	Binary Output (Transistor, 40x1, 24-60VDC)	optional
DO-2210/PCCO2X	Checked command output 24-60 VDC	optional
AI-2300/PASI25	Analog in-/output (16x ± 20 mA + 4x2 opt. expans.)	optional
AI-2301/TEMP25	Analog input (32x PT-100, Ni-100)	optional
MX-2400/USIO2X	Signal Input/Output (24-60VDC, ± 20 mA, opt. exp.)	optional

Peripheral Elements TM 1703 ACP

Type	Designation	
PE-6410/USIO66	Peripheral Controller (Ax-PE bus el)	optional
PE-6411/USIO66	Peripheral Controller (1x Ax-PE bus opt)	optional
PE-6412/USIO66	Peripheral Controller (2x Ax-PE bus opt)	optional
PE-6410/TCIO66	Peripheral Controller for TC 1703 (Ax-PE bus el)	optional
PE-6411/TCIO66	Peripheral Controller for TC 1703 (Ax-PE bus opt)	optional
PE-6412/TCIO66	Peripheral Controller f. TC 1703 (2xAx-PE bus opt)	optional

1.2.2. Replaced Modules

Type	Discontinuation	Termination	Replacement ersatz
PS-6620	30.09.2008	30.09.2010	PS-6630
PS-6621	30.09.2008	30.09.2010	PS-6630
CP-2012	31.08.2008	31.12.2008	CP-2017
SM-2541	30.09.2008	30.09.2010	SM-2551
SM-2541	30.09.2008	30.09.2010	SM-2556 + SM-0551
SM-2542	01.01.2008	30.06.2008	SM-2556
SM-2554	01.10.2007	31.03.2008	SM-2556
CM-6830	01.04.2008	30.09.2008	CM-0843
PE-6400	30.09.2008	30.09.2010	PE-6410
PE-6401	30.09.2008	30.09.2010	PE-6411
PE-6402	30.09.2008	30.09.2010	PE-6412

**Note**

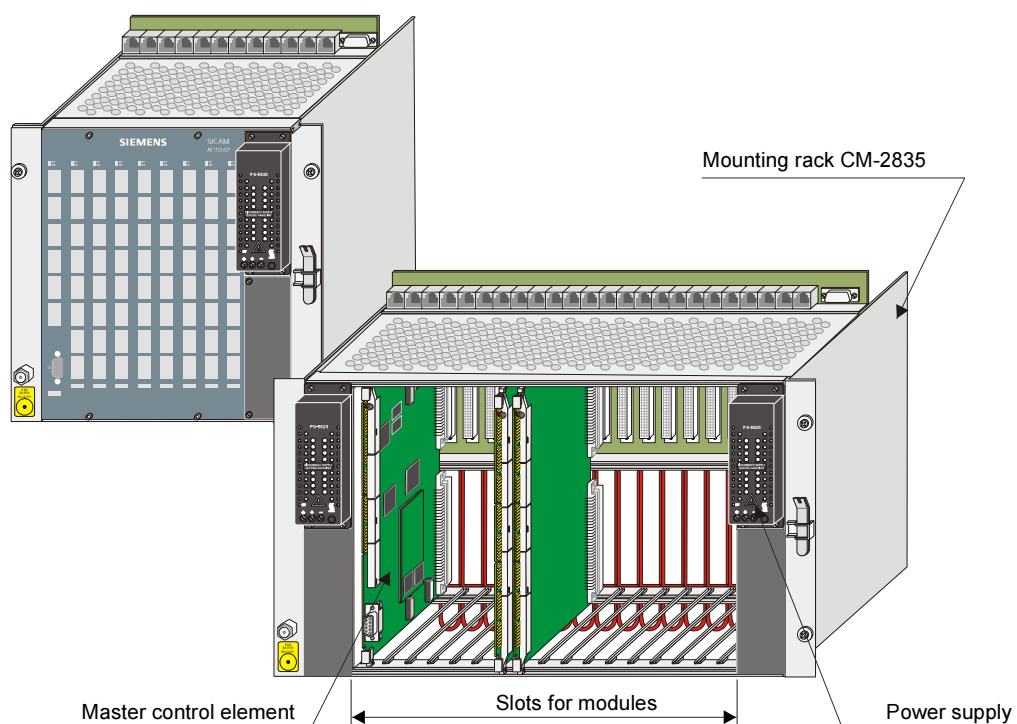
When replacing a module with another type, the parameterization of the system must be adapted correspondingly.

1.2.3. System Design

Due to its modular design, AK 1703 ACP offers a wide variety of configuration options. You can find details thereto in the manual *ACP 1703 Platforms Configuration Automation Units and Automation Networks*.

In AK 1703 ACP two mounting racks are available:

Type	Designation
CM-2832	AK 1703 ACP Mounting Rack with 9 Slots
CM-2835	AK 1703 ACP Mounting Rack with 17 Slots



1.3. Engineering PC

1.3.1. Required Hardware

For the engineering of AK 1703 ACP fits a PC with a serial (and optional ethernet interface) that fulfills the [Preconditions For The Operation Of The TOOLBOX II](#).

You will find recommended PC products in the continuously updated [PC Products Preference List](#). Should that be not at your disposal, please consult your contact person at Siemens.

1.3.2. Required Software

1.3.2.1. TOOLBOX II

The software is available as DVD and essentially consists of 4 Toolsets. Information and updates concerning these Toolsets can be found in *PRISMA*.

Toolset	Designation	Item number / MLFB
EM II	Engineering Manager	V30-016 (6MF18030AA160AA0)
PSR II	Configuring And Service Computer	V30-017 (6MF18030AA170AA0)
OPM II	Object Oriented Process Data Manager	V30-004 (6MF18030AA040AA0)
CAEX <i>plus</i>	Tool For Open/Closed-Loop Control Functions	V30-090 (6MF18031AA000AA0)

You find the guidance for handling updates in the booklet of the TOOLBOX II DVD.

Firmware For The System Elements

Firmware files are available in *PRISMA*. You can find this under `<item_number>/XX/53*`.

Examples:

System element	Designation	Item number
CP-2010/CPC25	MC25 Central System Function and Comm.	SC2-025-1
	TU MC25 TB II-Update	SC2-025-1.XX/53
	PC25 Ax 1703-Periph. Interf. and Autom.	SC2-026-1
	TU PC25 TB II-Update	SC2-026-1.XX/53
MX-2400/USIO26	USIO26 Universal Signal In-/Output IEC	SA2-426-1
	TU USIO26 TB II-Update	SA2-426-1.XX/53
PE-6400/USIO65	Peripheral Controller (Ax-PE bus el)	SC6-401-1
	TU USIO65 TB II-Update	SC6-401-1.XX/53

1.3.2.1.1. Preconditions For The Operation Of The TOOLBOX II

Prerequisite for the operation of the TOOLBOX II is the utilization of an appropriate PC, that must fulfil certain hardware requirements, depending on the license pack you purchased. You find further information thereto in the booklet of the TOOLBOX II DVD.

Preconditions For The Installation Of The TOOLBOX II

- DVD drive
- Windows standard administrator rights (read and write rights for the file system and the Windows Registry)
- NTFS partition
- Installed TCP/IP
- Other ORACLE applications must not be installed on the computer (database, client, etc.)
- A C:\ drive must exist
- Necessary Windows Regional Options:
 - 'Dot' for decimal separating character and 'comma' for the thousands separating character must be different
 - No 'single inverted comma' may be used as separating character
- Use of Windows Fonts in normal size (96 DPI)
- Microsoft Internet Explorer as of version 6.0
- Adobe Flash Player as of Version 7.0 (to play online help tutorial videos), see Installation DVD directory \MACROMEDIA or www.adobe.com
- An initial installation needs 3.5 GB free harddisk memory
- For an update, an additional 3 GB of temporary hard disk memory is required

Supported Operating Systems (32 Bit Platform)

Configuration	Operating system
TOOLBOX II Local	Windows XP Professional (as of Service Pack 2)
TOOLBOX II Server	Windows 2003 Server (as of Service Pack 2)
TOOLBOX II Web Server	Windows 2003 Server (as of Service Pack 2) with Terminal Services

For the use of TOOLBOX II, user rights of the group "User" under Microsoft Windows are required.

Recommended Hardware Requirements

Configuration	RAM	Disk	CPU
TOOLBOX II PC	1 GB	40 GB	Pentium 4 2 GHz
Notebook	2 GB	80 GB	Core 2 Duo 1.8 GHz
Workstation	3 GB	160 GB	Core 2 Duo 2.4 GHz
Server	4 GB	2 x 250 GB (RAID)	XEON-5160 3 GHz

1.4. Basics For Engineering

Before you begin with the engineering, the predefined configuration parameters of the TOOLBOX II are to be checked and if necessary changed (refer to section [2.2, Presets For The Engineering](#)).



Note

All operation and test functions of the AK 1703 ACP described in this manual are generally applicable for the "Administrator" role available in the TOOLBOX II.

1.4.1. Storage Of Engineering Data

Engineering data maintained with the TOOLBOX II are stored differently during the engineering process. The TOOLBOX II generally works with data, that are stored in a database on the hard disk of the Toolbox PC.

Merely loading operations are able to load engineering data from the hard disk of the Toolbox PC into a target system. There, they are stored in the flash card of the master control element.

With startup of the target system, data that differ from that in the flash PROM, are transferred to the flash PROM.

1.4.2. Offline Engineering

With offline engineering there is no physical connection to the target system. The engineering data are stored in a file on the Toolbox PC.

Activities With Offline Engineering (Recommended Sequence)

- Presets
- Configure plant
- Configure assembly
- Parameter setting system
- Parameter setting telecontrol functions
- Edit function diagram
- Simulate function diagram
- Import, export, backup of engineering data
- Documentation of the hardware
 - HW configuration
 - Pin assignment
- Documentation of the software
 - Parameters
 - Function diagram

1.4.3. Online Engineering

With online engineering the target system is connected with the engineering PC and switched on. In the master control module of the target system a flash card must be inserted.

Essential Activities With Online Engineering

- Load system parameters
- Load telecontrol parameters
- Load application program
- Data flow test
- Message simulation
- Topology test
- Function diagram test
- Diagnostic
- Revision interrogation

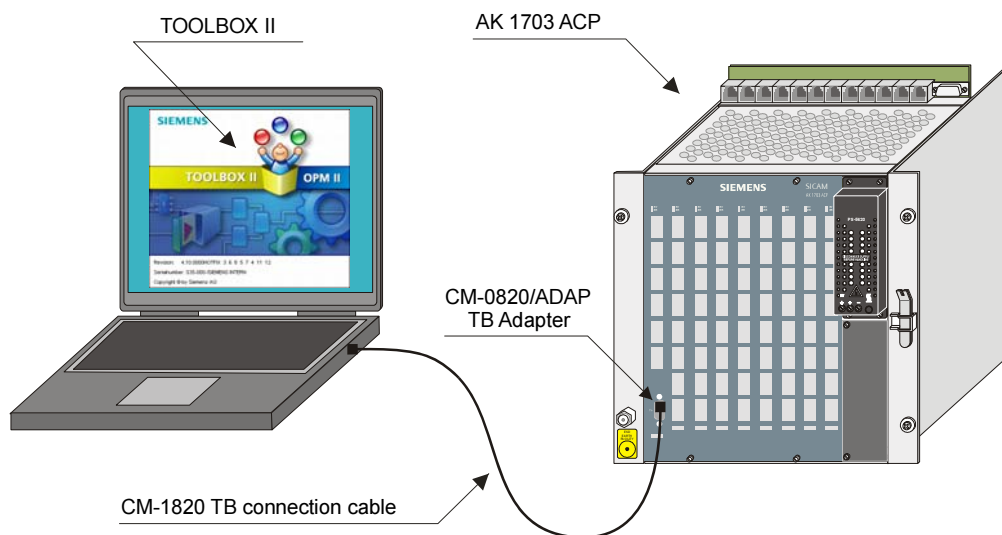
1.4.3.1. Transfer Of Engineering Data To AK 1703 ACP

The parameter setting of AK 1703 ACP is possible online via the following connections:

- direct local via the serial Toolbox interface (TB) on the master control element
- remotely
 - serial via modems
 - Ethernet (TCP/IP)
 - Ethernet (TCP/IP) and Terminal Server (serial)

1.4.3.1.1. Direct Connection

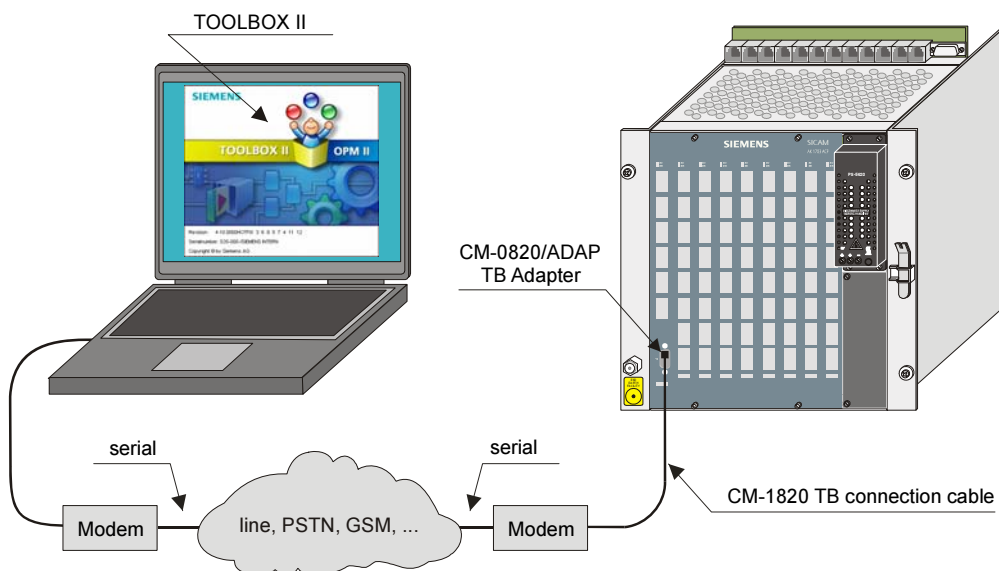
Toolbox PC and AK 1703 ACP are connected directly with the Toolbox cable.



1.4.3.1.2. Remote Connection

Serial Via Modem

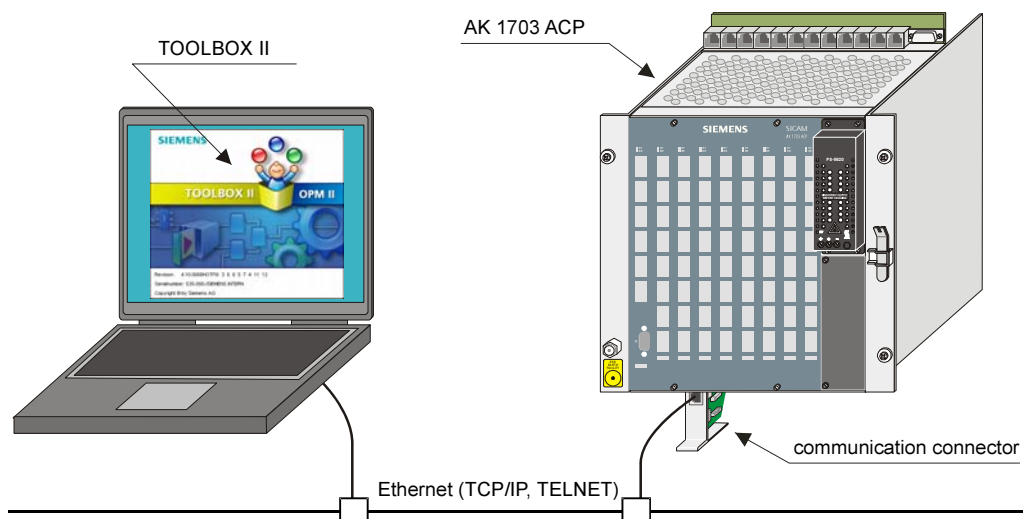
Toolbox PC and AK 1703 ACP are connected via 2 serial modems.



Ethernet (TCP/IP)

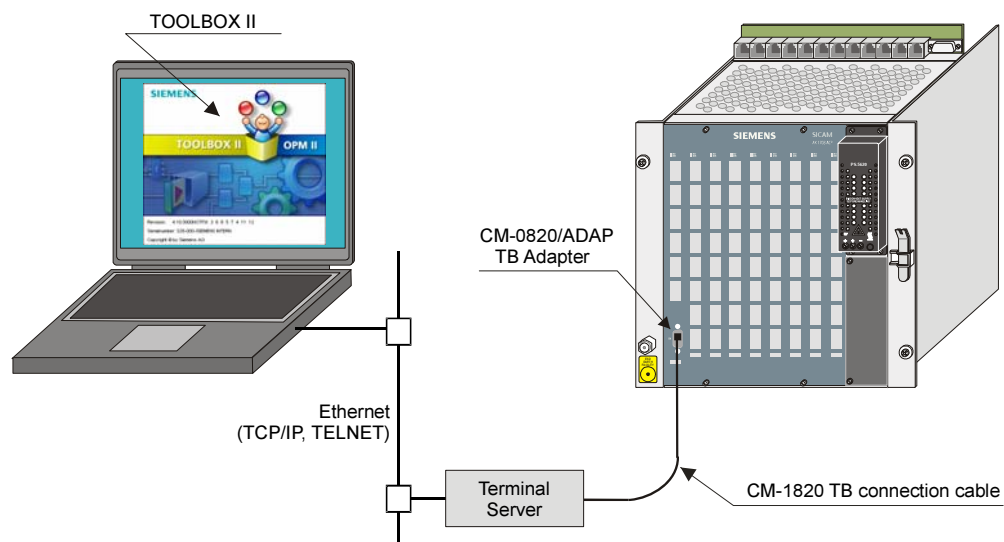
Toolbox PC and AK 1703 ACP are connected via Ethernet (TCP/IP).

For this, the master control element or the processing- and communication element must be equipped with a protocol element for Ethernet (refer to section 3.1.3, [Configure Automation Unit With System Elements](#)). The Ethernet protocol must be initialized and set. Further information can be found in the manual *AK 1703 ACP Installation*, section "Setup of External Communication".



Via Ethernet (TCP/IP) And Terminal Server (Serial)

Toolbox PC and AK 1703 ACP are connected via Ethernet (TCP/IP) and Terminal Server (serial).



1.4.3.2. Logical Connection Between Toolbox PC And AK 1703 ACP

With engineering, in each case a logical connection must exist to that automation unit, that is the object of the engineering task. It is differentiated between:

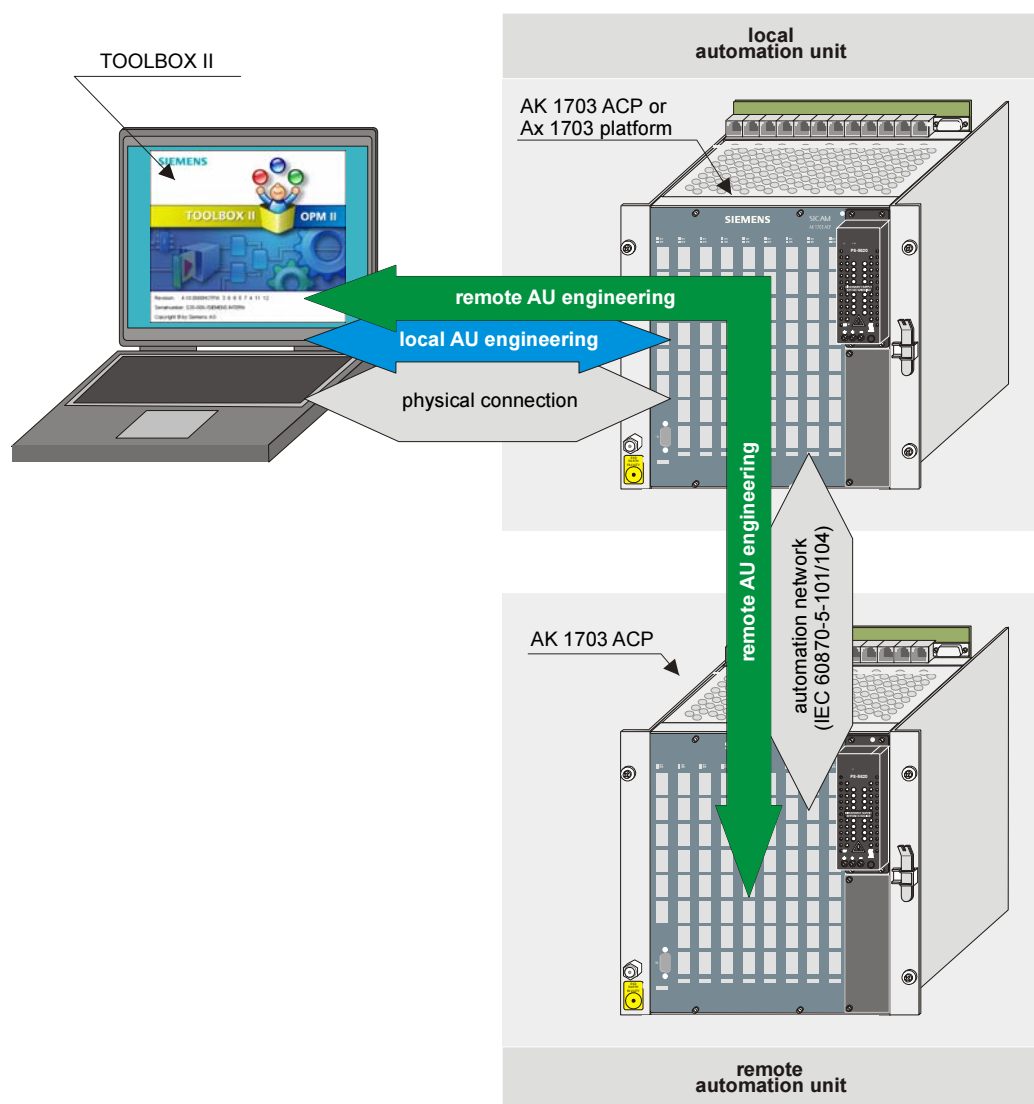
- Engineering of a local automation unit
(that is that one, to which a physical connection exists, regardless of whether direct or remote)



Attention

A first initialization of an automation unit is only possible in this way.

- Engineering of a remote automation unit
(automation unit, that can be reached via the local automation unit; a consistent remote communication according to IEC 60870-5-101 or -104 is required)



1.5. Switching The System On And Off

1.5.1. Switching On

Before switching the system on, all system elements must be connected to a power supply. Switching on takes place by connecting the voltage, as for example by switching on a miniature circuit breaker. The system starts up automatically (startup after power-up).

The entire system is operational (without consideration of the error display), as soon as all available system elements are supplied with power and all system elements have concluded the startup (refer to section [4.2, Checks And System Displays](#)). This is also applicable analogously, if only parts of the system are switched off and switched on again.

1.5.2. Switching Off

The switching off of the system takes place by disconnecting the voltage on all system parts. If the power is switched off only at the master control element, not autarchic subordinate basic system elements are not operational.

Caution

Autarchic subordinate basic system elements form together with their subordinate system elements a subsystem, that – as far as supplied completely with voltage – despite of disconnecting the voltage on the master control element remains entirely operative.

Caution

The switching off of the master control element during writing operations to the flash card (load firmware, load parameters) is to be absolutely avoided, since the data on the flash card could be destroyed as a result.

2. Administration

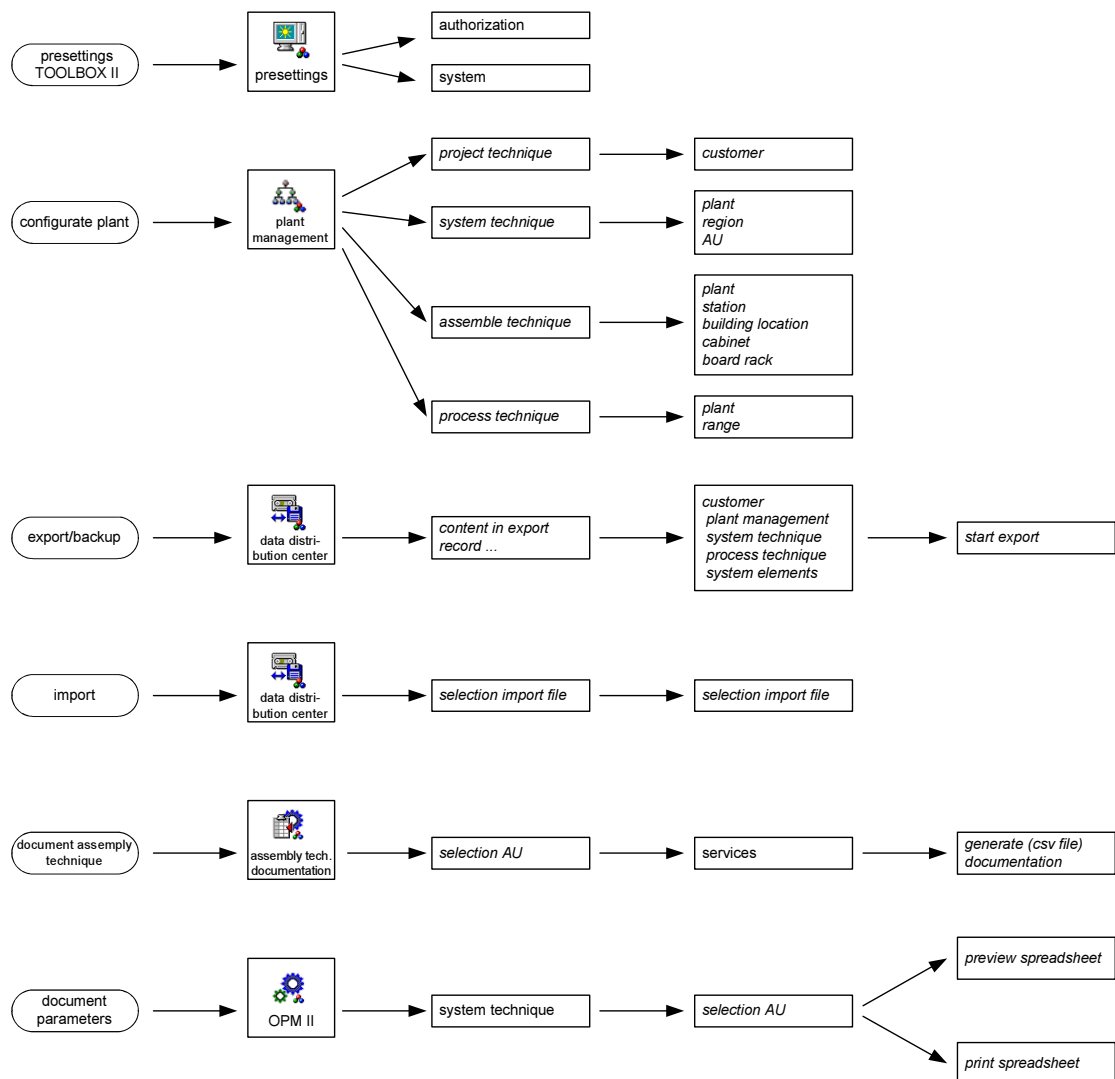
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2.1. General

The following section gives an overview of the preparing activities for the engineering of plants, as well as the functions for documentation and storage of the engineering data.

Overview Of The Essential Administrative Functions



2.2. Presets For The Engineering

Before you begin with the engineering of AK 1703 ACP, several basic settings are to be performed for the work with the TOOLBOX II:

- User and access rights
- User-specific settings
- Workplace-specific settings

These globally valid configuration parameters in the TOOLBOX II are displayed and set with the tool "TOOLBOX II Presets". They can - dependent on the access rights - be changed at any optional time.

You find the instruction thereto in the *TOOLBOX II User Manual*, chapter "*TOOLBOX II Presets*".

2.2.1. Notes about Users and Rights

Names of users (max. 8 characters) and roles (max. 20 characters) can be freely assigned. Each user can be assigned a special role.

For each role, certain rights can be freely selected and assigned from a list. An exception are the unchangeable roles, that are reserved for the SIEMENS specialists for maintenance purposes.

Depending on which rights a user has been assigned, it is determined which functions he may control.

You find the instruction thereto in the *TOOLBOX II User Manual*, chapter "*TOOLBOX II Presets*", section "*User/Role Administration*".

2.3. Configure Plant

The plant configuration is entered centrally in the TOOLBOX II with the tool "Plant Management" and subsequently complemented, changed or deleted.

The configuration data are parameterized in each case specifically for the various perspectives of the plant:

- Project technique
- Assembly technique
- System technique
- Process technique

You find the instruction thereto in the *TOOLBOX II User Manual*, chapter "Plant Management".



Note

The [initial creation](#) of a plant can be accomplished easily by means of "Wizards" with the tool "OPM II".

2.4. Import, Export, Backup of Engineering Data

With the tool "Data Distribution Center" the importing and exporting of parameters, as well as the creation of backup copies is enabled.

You find the instruction thereto in the *TOOLBOX II User Manual*, chapter "*Data Distribution Center*".

3. Engineering

Contents

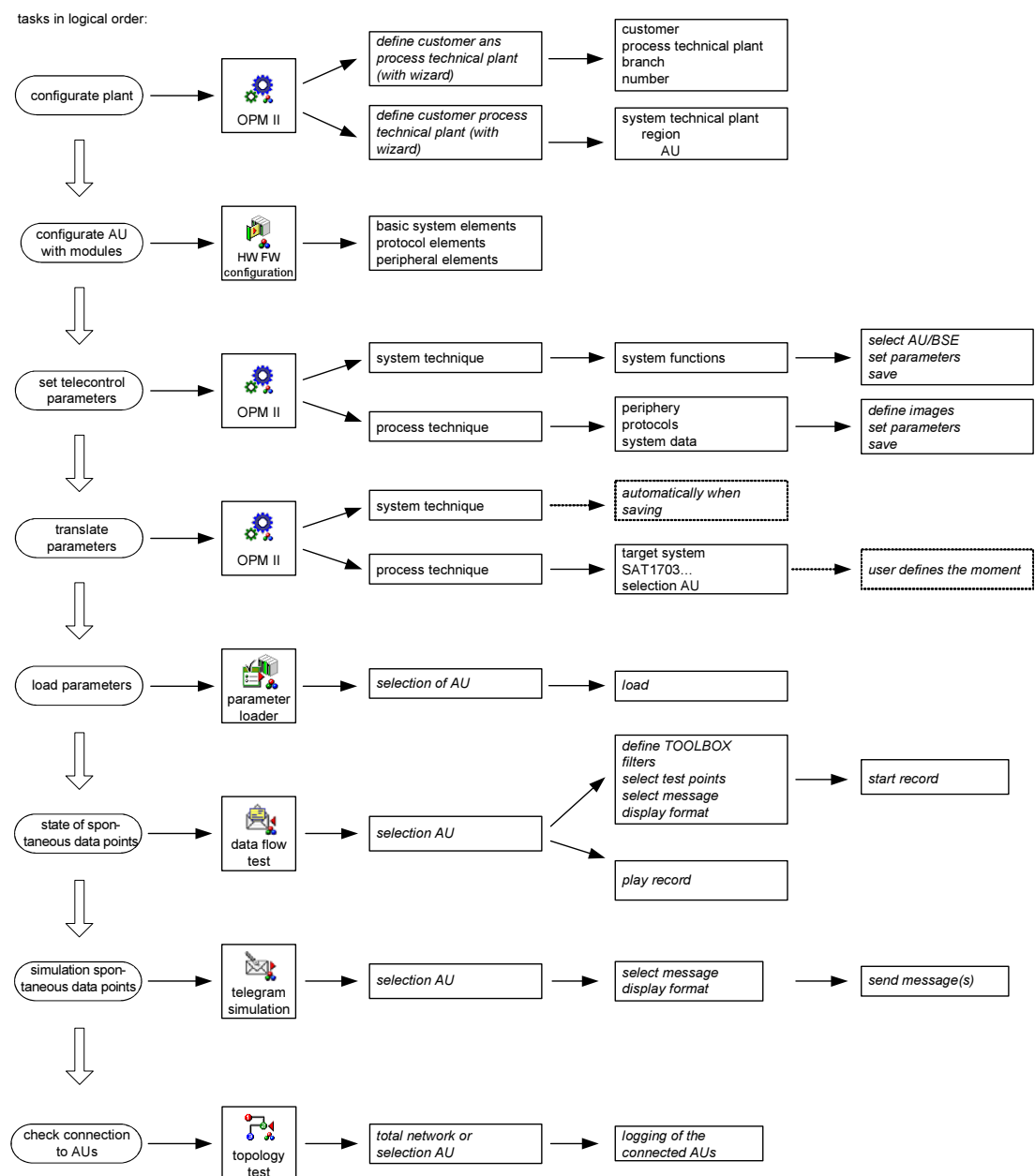
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3.1. Telecontrol

The acquisition, description, parameter setting and documentation of the process-technical process and its associated data points for the automation system is enabled with the TOOLBOX II, tool "OPM II" (Object Orientated Process Data Manager). The detailed description thereto can be found in the *TOOLBOX II User Manual, Chapter "OPM II"*.

The description of the function package Telecontrol can be found in the manual *SICAM 1703 Common Functions System and Basic System Elements, chapter "Telecontrol"*.

Overview Of The Fundamental Procedure For The Parameter Setting



3.1.1. Starting The TOOLBOX II

With initial start of a tool of the TOOLBOX II you must enter a user name and a password (refer to section [2.2, Presets For The Engineering](#)). After that you can begin with the parameter setting.

The user logout can be carried out with the tool "Presets TOOLBOX II" through the menu AUTHORIZATION | LOGOUT. You can continue to operate tools that are still active, but the activation of tools requires another login.

If all tools are quit without logout, the user will remain logged in unless a logout from the Toolbox PC (or Toolbox Server) takes place.

3.1.2. Configure Plant

For the initial creation of a plant you must enter the configuration data into the TOOLBOX II (tool "OPM II"). This task is supported by "Wizards". The set configuration data determines the plant topology.

For the process-technical plant you must define levels. You find the instruction thereto in the *TOOLBOX II User Manual, chapter "OPM II", section "Levels"*.

The administration of the plant configuration is carried out with the tool "Plant Management" (refer to section [2.3, Configure Plant](#)).

3.1.3. Configure Automation Unit With System Elements

With the tool "HW/FW Configuration" you can configure the single slots of an automation unit with system elements:

- Master control element
- Processing and communication elements
- Protocol elements
- Peripheral elements

The hardware configuration forms the prerequisite for the parameterization of the telecontrol function and of the automation function.



Note

With usage of TM 1703 ACP periphery no individual I/O modules are configured, rather only entire peripheral elements.

With the tool "OPM II" the I/O module is specified by attaching a data point from the process technique to an I/O in the system technique (for example binary input module).

Information on the configurable system elements can be found in the respective system element manual.

The information concerning the entry of the setting values can be found in the *TOOLBOX II User Manual, chapter "HW/FW Configuration"*.

Below the level of the Ax peripheral bus the configured peripheral elements with the process-technical signals (I/Os), which have been set in the window EDIT IMAGE, are displayed:

- Hardware pins
- Software data points

By means of the pop-up menu of a hardware pin, function EDIT IMAGE, you directly reach the process-technical parameter setting.

The information concerning the entry of the setting values can be found in the *TOOLBOX II User Manual, chapter "OPM II", section "System Technique"*.

3.1.4.2. Process-Technical Settings

The parameter setting takes place via the menu item TOOLS | IMAGES via the navigation tree of the process-technical plant. To the levels of the process-technical plant you must create here the images (real objects of the plant).

For the simplification of the engineering of large quantities of objects, parameters and values, different types can be defined below the levels:

- User types
- Link types
- Info types
- Parameter types

The definition of process-technical signals (I/Os) and their parameters for the technological processing takes place via the navigation tree in the window EDIT IMAGE, below the selected station:

- Parameters of the periphery
 - Addressing of the I/Os
 - Parameters for the preprocessing of signals
 - Parameters for the postprocessing of signals

The assignment of a process-technical signal to a hardware pin or software data point takes place in the system technique (M-CPU) by means of assignment of the link images.

The detailed instruction for the process-technical parameter setting can be found in the *TOOLBOX II User Manual, chapter "OPM II", sections "Images", "Types", "Assign"*.

The description of the technological processing of inputs and outputs (process peripheral signals) can be found in the manual *SICAM 1703 Common Functions Peripheral Elements according to IEC 60870-5-101/104*.

The description of the parameters can be found in the *TOOLBOX II User Manual, chapter "Parameter Documentation"*.

3.1.5. Transform Parameters

Before loading into the target system, the set parameters of the process-technical plant must be transformed. This is carried out with the tool "OPM II" through selection of the function DESTINATION SYSTEMS | SICAM 1703... | SICAM 1703 TRANSFORMER.

System-technical parameters are automatically transformed when saved.

Further details can be found in the *TOOLBOX II User Manual*, chapter "OPM II", section "Transform and Load", "CAEx plus".

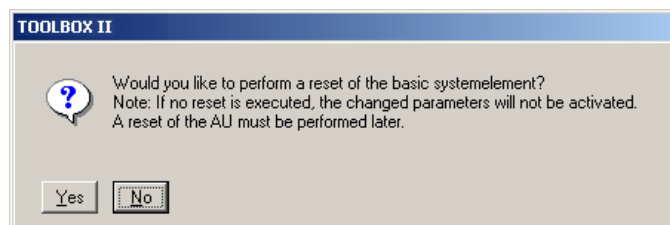
3.1.6. Upload Parameters

The loading of the parameters of a process-technical plant from PC to the target system takes place with the tool "Load parameters".

All parameter settings, that have been performed in the TOOLBOX II, are thereby stored together on the flash card in the master control element:

- Configuration parameters
- System parameters
- Process-technical parameters

If a changed parameter requires a startup of the basic system element, then a notice appears in the TOOLBOX II, as soon as the parameter has been accepted by the target system. The user determines, whether the startup is to be performed immediately or at a later time – for instance if further changed parameters are to be activated jointly.



Attention

During a loading operation, the switching off of the master control element is to be absolutely avoided, since the data on the flash card could be destroyed as a result.

For the loading of the parameters there are different variants available. You find the instruction thereto in the *TOOLBOX II User Manual*, chapter "Parameter Loader".

3.1.7. Parameter Comparison

With the parameter comparison function it can be ascertained whether the parameter status in the destination system is current. The comparison can be started with the tool "OPM II" via the context menu of the automation unit (right mouse button, PARAMETER | COMPARISON).

In the error output window of the "OPM II" appears the display whether the parameters of the selected automation unit are current or not current.

3.1.8. Test Functions

3.1.8.1. Status Of Spontaneous Data Points

With the tool "Data Flow Test" data streams (flow of messages) can be simultaneously logged and visualized. For a subsequent analysis - also at another location - a recording can take place with the Interface Recorder.

The function SIMULTANEOUS-LOG. serves for the recording of messages within the internal data flow of an automation unit. In protocol elements, the data flow from and to other automation units can also be acquired.

The following data can (with change of the spontaneous data) be simultaneously logged:

- Change of state of inputs/outputs
- Communication from and to protocol elements
- Data traffic from and to the application program
- Data traffic from and to special functions (for example set counters, set time)

By means of triggering a General Interrogation all spontaneous input signals can be simultaneously logged at any arbitrary time.

The data flow test provides extensive filtering options. You find the instruction thereto in the *TOOLBOX II User Manual, chapter "Data Flow Test"*.

Further details can be found in the manual *ACP 1703 Common Functions System and Basic System Elements, chapter "System Services", section "Data Flow Test"*.

3.1.8.2. Simulation Of Spontaneous Data Points

With the tool "Message Simulation", messages can be transmitted from the TOOLBOX II to automation units. Just one message or up to 100 messages in succession can be transmitted; in addition sequential delays and message repetitions can be defined.

The messages can be passed in at defined points of the system. With this function the following possibilities exist:

- Setting of outputs
- Simulation of the communication from and to protocol elements
- Data traffic from and to the application program
- Data traffic from and to special functions (for example set counters, set time)

It is to be observed, that due to the simulation the system state can change.



Warning

The manual controlling of outputs with the plant running can lead to damage to persons and machines. Ensure that aggregates in the control area of the command output and those subsequent aggregates in the process chain are protected and that persons in the vicinity are warned.

You find the instruction thereto in the *TOOLBOX II User Manual, chapter "Message Simulation"*.

3.1.8.3. Check The Connection To Automation Units

The tool "Topology Test" is used for the acquisition of all automation units in a SICAM 1703 automation network that are reachable or non-reachable from the automation unit momentarily physically connected.

You find the instruction thereto in the *TOOLBOX II User Manual, chapter "Topology Test"*.

3.1.9. Documentation

3.1.9.1. HW Configuration

The documentation of the automation unit slots occupied with system elements can be created with the tool "HW configuration". The output takes place optionally as preview on the screen or to a printer.

3.1.9.2. Assembly Technique

With the tool "Assembly-technical documentation" the assignment of all hardware pins within an assembly-technical plant can be documented. The output takes place to a text file as a table (format *csv*):

- All automation units
- Selected automation units
- All system elements of an automation unit
- Selected system elements of an automation unit

The following information is displayed:

- Physical structure of the system element
- System-technical address of each pin within a message
- Process-technical address of each pin within a message
- General information about a pin (long text)
- Assignment of the pin to a link image in OPM II

You find the instruction thereto in the *TOOLBOX II User Manual, chapter "Engineering Manager EM II", section "Assembly Technique Documentation"*.

3.1.9.3. Interface to Elcad

For the coupling with the design tool ELCAD, it is possible to transfer images that are attached to a AK 1703 ACP system element via this defined interface. The output takes place to a text file (format *asc*).

The generation of the file takes place with the tool "OPM II" via selection of the function DESTINATION SYSTEMS | SICAM 1703... | SICAM 1703 TRANSFORMER.

You find the instruction thereto in the *TOOLBOX II User Manual*, chapter "OPM II", section "Elcad".

3.1.9.4. Telecontrol Function

The documentation of the system-technical and process-technical parameters can be created with the tool "OPM II". The output takes place optionally as preview on the screen or to a printer:

- System-technical parameters
 - Window SYSTEM TECHNIQUE
Select an automation unit or a master control element (M-CPU), or a subordinate group of parameters by means of clicking the right mouse button.
Selection PRINT DOCUMENTATION or PREVIEW DOCUMENTATION.
- Process-technical parameters
 - Window EDIT IMAGE
Select an image by means of clicking the right mouse button.
Select PRINT PREVIEW or PRINT.
 - Window BULK EDIT
Documentation of the current content (process-technical addresses and parameters).
Select menu item SYSTEM | PRINT PREVIEW or SYSTEM | PRINT.

You find the instruction thereto in the *TOOLBOX II User Manual*, chapter "OPM II", section "General", section "Documentation".

3.2. Automation

For the management of automation tasks the *open-/closed-loop control function* is used. This is freely definable with the help of an application program.

The application program processes process information from the peripheral elements connected to the basic system element and/or from other system elements in the automation network of the specific process-technical plant.

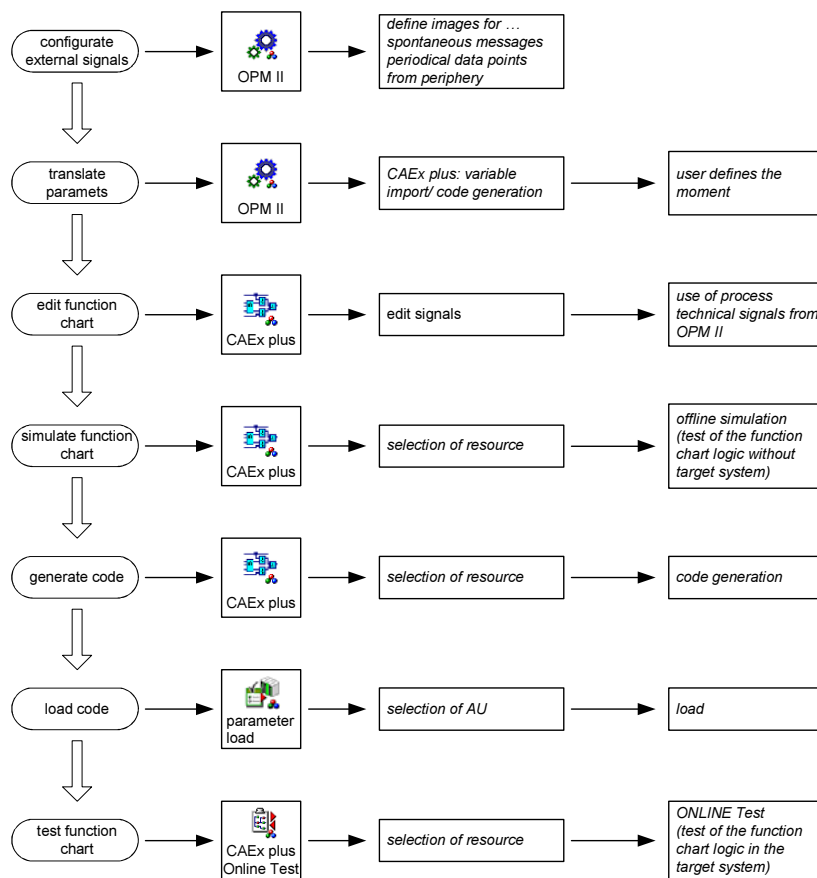
The creation of the application program is enabled with the TOOLBOX II, tool "CAEx plus". The detailed description thereto can be found in the *TB II CAEx plus User Manual*.

The tool "CAEx plus" supports the programming languages "FBD" (function block language, function diagram) and "SFC" (Sequential Function Chart, sequence language) according to the standard IEC 61131-3 (Programmable Logic Controllers Part 3: Programming Languages).

The description of the function package Telecontrol can be found in the manual *SICAM 1703 Common Functions System and Basic System Elements, chapter "Automation"*.

Overview Of The Fundamental Procedure For The Programming

tasks in logical order:



3.2.1. Configuring External Signals

The signals of the process-technical plant created with the "OPM II" can be used by CAEx *plus* after the transformation (refer to section [3.1.5, Transform Parameters](#)). The structure of the project tree automatically adapts to the structure from the OPM II.

An introduction how to create a project can be found in the *TB II CAEx plus User Manual*, chapter "First Steps" and chapter "Additional Products", section "Signal List (Optional)", as well as in the Online Help of "CAEx plus".

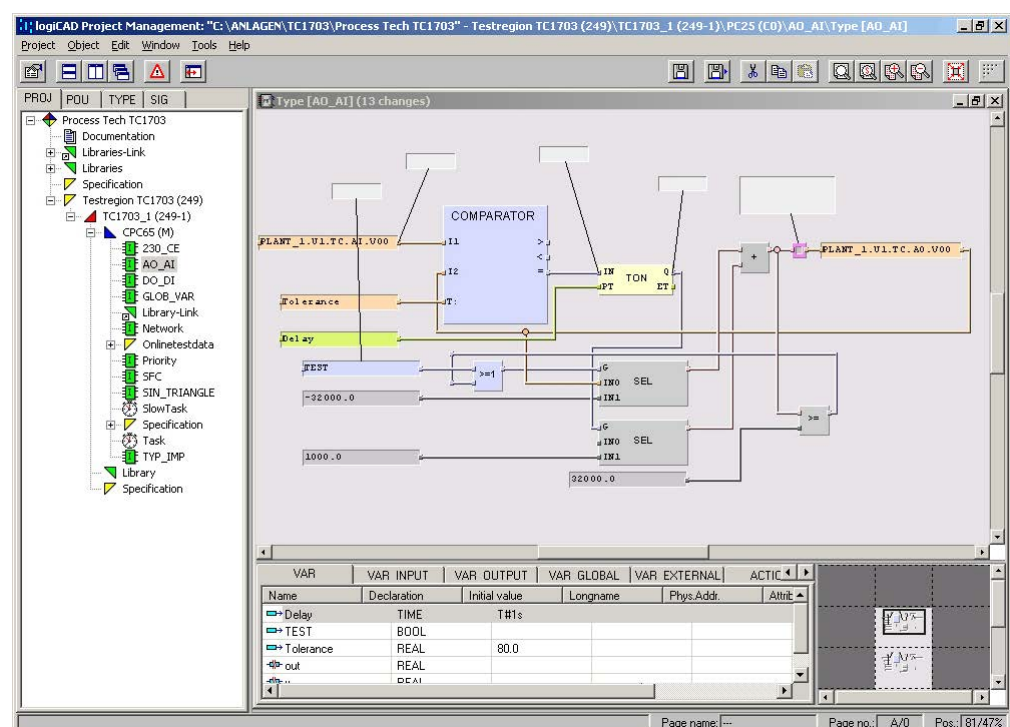
3.2.2. Create Function Chart

The tool "CAEx *plus*" provides various editors and standard libraries for the creation of the open-/closed-loop control functions.

The process-technical functions of a plant are created with the function diagram editor (FBD-Editor). A function diagram is thereby created by the interconnection of

- predefined functions and function blocks ("CAEx *plus*" standard library)
- functions and function blocks defined by the user

The exact description of the editors is included in the *TB II CAEx plus User Manual*, chapter "Editors", as well as in the Online Help of "CAEx *plus*".



Additional Information can be found in the *TOOLBOX II User Manual*, chapter "CAEx *plus*".

The technical details for the processing of the open-/closed loop control functions and their subfunctions can be found in the manual *ACP 1703 Common functions System and Basic System Elements*, chapter "Automation".

3.2.3. Simulate Function Chart

A Function Diagram can be tested in "CAEx plus" with the OFFLINE SIMULATION.

This function can be called via the context menu of the CPU, program instance or type instance (right mouse button). You can find the details thereto in the *CAEx plus User Manual*, chapter "Additional Products", section "Offline Simulation".

3.2.4. Generate Code

Before the loading of the open-/closed-loop control functions into the target system, the application program code must be generated.

The code generation can be started via the CAEx plus Transformer or from "CAEx plus" via the context menu of the CPU (right mouse button).

The function diagram is now translated and checked by a compiler. Detected errors are output in detail after the code generation in the CAEx plus "Error-state viewer".

In addition, information about the occupied storage space is output:

- Currently used memory for code (kB), load factor (%)
- Currently used memory for variables (kB), load factor (%)

Example For CAEx plus Code Generator Report

The screenshot shows the 'Error-state viewer' window in the CAEx plus software. The window displays a list of messages generated during the code generation process. The messages are organized by date and time, and include details about the code generation process for the 'AK 1703 ACP' project. The messages include information about matching data, source code generation, and memory usage.

Date/Time	Level	Text
20.10.2004, 12:00:43	Information	MSG010: Code generation for <C:\SAT\TBII\TEMP\vie_maa\CAExplus\TMP-2005_ANL-00000001\AK 1703 ACP L2P\VE [249] L2F\AK1703 test [249.0] L2C\PC
20.10.2004, 12:00:43	Information	MSG012: Matching data started
20.10.2004, 12:00:43	Information	MSG013: Matching data finished
20.10.2004, 12:00:47	Information	MSG001: Source code generation started for <Calc>
20.10.2004, 12:00:47	Information	MSG002: Source code generation finished for <Calc>
20.10.2004, 12:00:54	Information	----- BEG. of VAR_EXTERNAL resolving -----
20.10.2004, 12:00:54	Information	number of input-telegrams for the high-prior task = 0 (maximum number = 100)
20.10.2004, 12:00:54	Information	number of input-telegrams for the middle-prior task = 1 (maximum number = 2048)
20.10.2004, 12:00:54	Information	number of input-telegrams for the lower-prior task = 0 (maximum number = 2048)
20.10.2004, 12:00:54	Information	number of output-telegrams without threshold = 0 (maximum number = 3072)
20.10.2004, 12:00:54	Information	number of output-telegrams with threshold (measure value telegram) = 0 (maximum number = 1024)
20.10.2004, 12:00:54	Information	----- END. of VAR_EXTERNAL resolving -----
20.10.2004, 12:00:55	Information	MSG003: Binary code generation started for <PC25 (C0)>
20.10.2004, 12:01:03	Information	----- Complex messages start here -----
20.10.2004, 12:01:03	Information	----- End of complex messages -----
20.10.2004, 12:01:03	Information	----- Link messages start here -----
20.10.2004, 12:01:03	Information	Currently used memory: code (27.60 kB - load factor: 5%) variables (4.30 kB - load factor: 0%)
20.10.2004, 12:01:03	Information	----- End of link messages -----
20.10.2004, 12:01:03	Information	----- CAEx-parameterblocks generated -----
20.10.2004, 12:01:03	Information	MSG004: Binary code generation finished for <PC25 (C0)>
20.10.2004, 12:01:04	Information	MSG018: Error=0/warnings=1
20.10.2004, 12:01:04	Information	MSG009: Error-free code generated for <C:\SAT\TBII\TEMP\vie_maa\CAExplus\TMP-2005_ANL-00000001\AK 1703 ACP L2P\VE [249] L2F\AK1703 test [249.0]

The exact description of this tool is included in the *TB II CAEx plus User Manual*, chapter "Target-System Connection", section "Functions for Target-System Connection", section "Code Generation".

The code generation can be started independently from "CAEx *plus*" with the tool "OPM II" through selection of the function DESTINATION SYSTEMS | CAEX PLUS... | SIGNAL LIST/GENERATE CODE. Thereby a signal list is created for the processing of the parameterized signals in the open-/closed loop control function and read into CAEx *plus*.

Further details can be found in the *TOOLBOX II User Manual*, chapter "OPM II", section "Transform and Load", "CAEx *plus*".

3.2.5. Load Code

Before a function chart can be loaded into the target system, it must be converted to an Instruction List (ASCII format) readable for TM 1703 ACP with the function "CAEx *plus* Transformer".

The loading of the application program code into the target system takes place jointly with the parameters, and can be started from the System Technique of the tool "OPM II" via the context menu of the respective automation (right mouse click, PARAMETER | LOADING).

All open-/closed-loop control functions, that have been performed in "CAEx *plus*", are thereby stored on the flash card in the master control element.

Another option offers the tool "Load parameters" (refer to section [3.1.6, Upload Parameters](#)).

The loading of the application program code normally takes place without operational interruption. Under certain conditions however a startup of the target system is required. In this case, after the compiling of the application program a corresponding warning is output in the "Error-state viewer" of "CAEx *plus*".

The technical description thereto can be found in the manual *ACP 1703 Common Functions System and Basic System Elements*, chapter "Automation", section "Loading of Application Program (Reload)".

Exact details for the operation can be found in the *TOOLBOX II User Manual*, chapter "Parameter Loader".



Attention

During a loading operation, the switching off of the master control element is to be absolutely avoided, since the data on the flash card could be destroyed as a result.

3.2.6. Test Function Chart

With the tool "CAEx *plus*", all open- and closed-loop technical tasks in processing elements of the AK 1703 ACP system can be tested online (selection of the CPU, function ONLINE-TEST in pop-up menu).

The following test functions are available:

- Display and force values
- Test switch input/output messages, input/output process images of the peripheral elements
- Changing the execution status of the open-/closed-loop control function
 - Stop controller
 - Start controller
 - Perform cold start of the resource
 - Perform warm start of the resource
 - Halt task
 - Continue task
 - Perform cold start of a task
 - Perform warm start of a task
 - Halt program
 - Continue program
- Setting breakpoints
- Real time archive
- Display status information
- Read and write variables
- Oscilloscope

The technical description of the online test function can be found in the manual *ACP 1703 Common Functions System and Basic System Elements, chapter "Automation", section "Online Test"*.

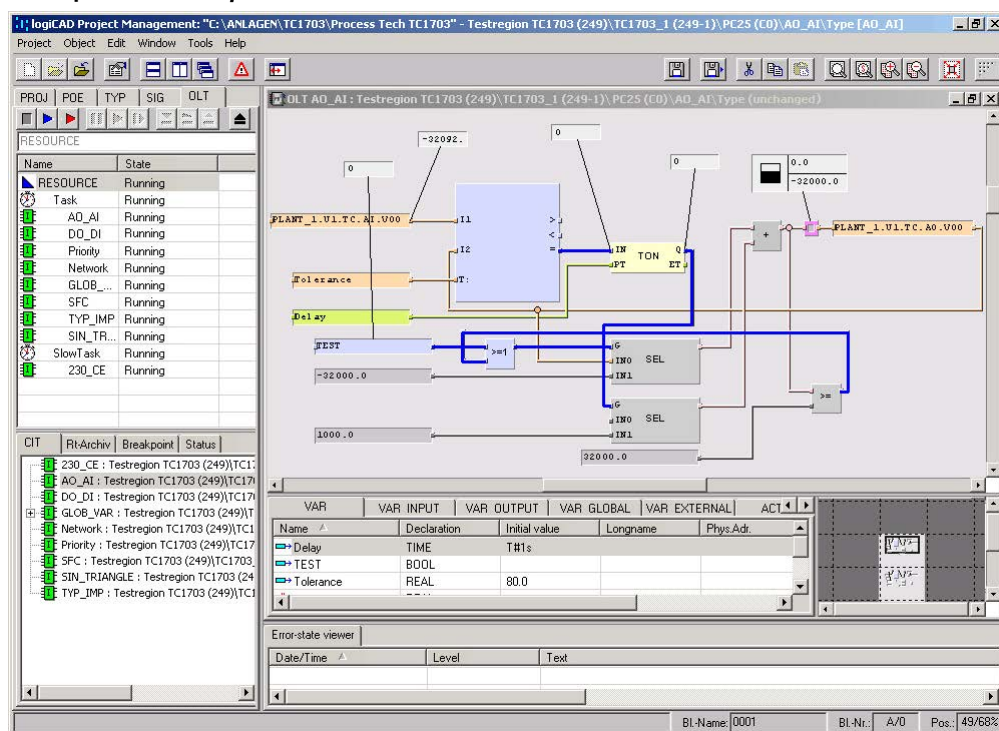
The details for operation can be found in the *TB II CAEx plus User Manual, chapter "Target-System Connection", section "Functions for Target-System Connection"*.

**Note**

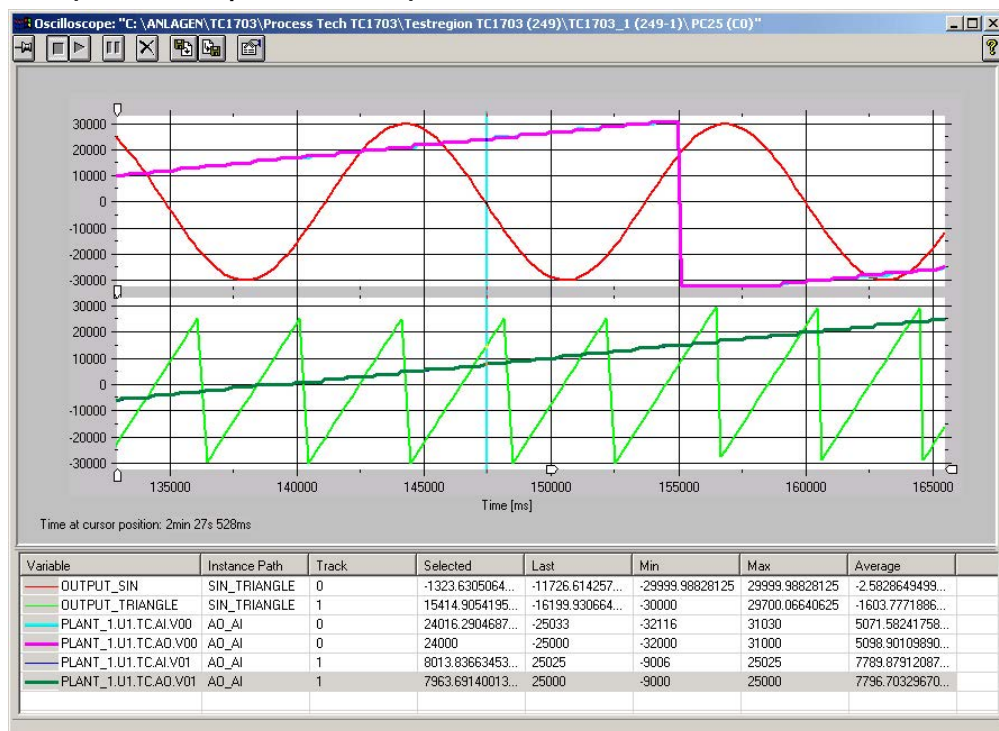
With user program running the simulated value of a variable is statically overwritten. In contrast to this, inputs/outputs are not statically overwritten, rather only with a change to their process image (edge-triggered).

With user program stopped, simulated values are retained.

Example For CAEx plus Online Test



Example For CAEx plus Oscilloscope



3.2.7. Documentation

3.2.7.1. Reference List

With the tool "CAEx *plus*" a reference list can be created via the project hierarchy, displayed on the screen or printed out. The reference list concerns alternatively

- CPU
- POU (Program Organization Unit)

and is executed by means of pop-up menu CROSS-REFERENCES.

You find the instruction thereto in the *TB II CAEx plus User Manual*, chapter "Basics", section "Default Operating Elements", section "Commands of the Pop-Up Menus".

3.2.7.2. Function Chart

In the project hierarchy of the tool "CAEx *plus*", by selecting one of each level, by means of pop-up menu the function PRINT can be executed. The print operation is started on the default printer of the PC.

You find the instruction thereto in the *TB II CAEx plus User Manual*, chapter "Basics", section "Default Operating Elements", section "Commands of the Pop-Up Menus", section "Print in Project Management".

Additional information can be found in the *TB II CAEx plus User Manual*, chapter "Basics", section "Default Properties" as well as chapter "Lists and References", section "Designing Printouts with DXF".

Further extensive possibilities for the documentation are provided by the optional function DOCUMENT MANAGEMENT (Additional product for CAEx *plus*).

You find the instruction thereto in the *TB II CAEx plus User Manual*, chapter "Additional Products", section "Documents Management (Optional)".

4. Maintenance

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4.1. Operation And Display Elements

Meaning Of The LED-Symbols

Symbol	State	Symbol	State
○	LED dark	⦿	LED flashes irregularly
●	LED lit	◐	LED flashes cyclically
◉	LED lights up briefly	X	not relevant

AK 1703 ACP supports the use of peripheral elements of the systems AK 1703 ACP, AM 1703 and TM 1703 ACP.

In the sections [4.1, Operation And Display Elements](#) and [4.2, Checks And System Displays](#) only system elements of the system AK 1703 ACP are suggested. Hints to system elements of other systems you will find in the corresponding documents.

4.1.1. Master Control Element, Processing And Communication Elements

LEDs on the front panel (refer to [A.1, Master Control](#)) display various states. The arrangement of the LEDs comprises

- System display elements
Display of operating- and error states of the system
- Function display elements
Display of the readiness of the protocol elements
- Communication display elements
Display of the operating- and error states of the protocol elements

Which states are displayed, is determined by

- the firmware implemented
- the protocol elements used

System Display Elements

Name	Color	Function	LED	Meaning
RY	yellow	Ready	●	Master control element ready
ER	red	Error	●	- Sum error (internal error, external error, warning, module failure, failure; including peripheral elements) - Startup
TST	yellow	Test	●	Test
WRN	yellow	Warning	●	Warning
BBD	yellow	Board breakdown	●	Module failure
INT	yellow	Internal error	●	Internal error
EXT	yellow	External error	●	External error
COM	yellow	Communication fault	●	Communication error
HLT	yellow	Halt	◐	Firmware shut down
			●	Module shut down
FLR	yellow	Firmware load request	●	PROM error
CAE	yellow	Computer aided engineering	◐	Toolbox operation remote
			●	Toolbox operation local
RUN	yellow	Run	○	- not used -
CPY	yellow	Copy	◐	Parameters are being loaded into the flash PROM
			●	Firmware being loaded
			⦿	Check checksum of parameter and firmware files on the flash card
SYN ^{*)}	yellow	Synchronization	⦿	Synchronization (minute pulse)
GI	yellow	General interrogation	⦿	General interrogation

^{*)} applies only for the master control element

Function Display Elements

Name	Color	Function	LED	Meaning
TSK0	yellow	Task 0	●	High prior task running
TSK1	yellow	Task 1	●	Middle prior task running
TSK2	yellow	Task 2	●	Low prior task running
M ACT ¹⁾	yellow	M active	●	M-CPU active
BSE ACT ²⁾	yellow	BSE active	●	Basic system element active
SI0 ACT	yellow	SI0 active	●	Protocol element interface 0 active
SI1 ACT	yellow	SI1 active	●	Protocol element interface 1 active
SI2 ACT ²⁾	yellow	SI2 active	●	Protocol element interface 2 active
SI3 ACT ²⁾	yellow	SI3 active	●	Protocol element interface 3 active
CO ACT ¹⁾	yellow	CO active	●	C0-CPU active

¹⁾ applies only for the master control element

²⁾ applies only for the processing and communication elements

Communication Display Elements For Serial Interface (SIP) And Network Interface (NIP)

Name	Color	Function ¹⁾	LED	Meaning
TX0 TX2 ²⁾	yellow	SIP: transmit data	●	Data are being transmitted
			●	Connection set up (dial-up traffic)
RX0 RX2 ²⁾	yellow	SIP: receive data	●	Data have been received
			●	
RT0 RT2 ²⁾	yellow	SIP: request to send	●	Send request to transmission facility
			●	
CD0 CD2 ²⁾	yellow	SIP: data carrier detect	●	Transmission facility has detected data carrier signal of another transmission facility
			●	
CT0 CT2 ²⁾	yellow	SIP: clear to send	●	Transmission facility ready to send (data can be transferred)
			●	
DT0 DT2 ²⁾	yellow	SIP: data terminal ready	●	Readiness for operation (message to transmission facility)
			●	
DS0 DS2 ²⁾	yellow	SIP: data set ready	●	Transmission facility ready for operation
			●	
ER0 ER2 ²⁾	yellow	SIP: error	●	Communication failure (all connections faulty)
			●	Communication fault (at least one connection faulty)
TX/LK1 TX/LK3 ²⁾	yellow	SIP: transmit data	●	Data are being transmitted
			●	Connection established (dial-up traffic)
RX/PK1 RX/PK3 ²⁾	yellow	NIP: ethernet link	●	Hardware connection to hub present
			●	
RT/SP1 RT/SP3 ²⁾	yellow	SIP: receive data	●	Data have been received
			●	Packet/frame to own station or broadcast received
CD/DP1 CD/DP3 ²⁾	yellow	SIP: request to send	●	Send request to transmission facility
			●	Transmission rate 100Mbps
CT1 CT3 ²⁾	yellow	NIP: speed	●	Transmission rate 10Mbps
			●	
DT1 DT3 ²⁾	yellow	SIP: data carrier detect	●	Transmission facility has detected data carrier signal of another transmission facility
			●	
DS1 DS3 ²⁾	yellow	NIP: Duplex	●	Full duplex
			●	Half duplex
ER1 ER3 ²⁾	yellow	SIP: clear to send	●	Transmission facility ready to send (data can be transferred)
			●	
DT1 DT3 ²⁾	yellow	NIP: -	●	Not used
			●	
DS1 DS3 ²⁾	yellow	SIP: data terminal ready	●	Readiness for operation (message to transmission facility)
			●	
DS1 DS3 ²⁾	yellow	NIP: -	●	Not used
			●	
ER1 ER3 ²⁾	yellow	SIP: data set ready	●	Transmission facility ready for operation
			●	
ER1 ER3 ²⁾	yellow	SIP: error	●	Communication failure (all connections faulty)
			●	Communication fault (at least one connection faulty)
ER1 ER3 ²⁾	yellow	NIP: error	●	Communication failure (all connections faulty)
			●	Communication fault (at least one connection faulty)

¹⁾ dependent on the configured protocol element

²⁾ applies only for the processing and communication elements

Communication Elements For Fieldbus Interface (FIP)

Name	Color	Function	LED	Meaning
RY0 ¹⁾ RY2 ²⁾	yellow	FIP: ready	●	Profibus ready
			◐	Bootstrap loader active
			⦿	Hardware or system error
			○	Hardware defect
RUN0 ¹⁾ RUN2 ²⁾	yellow	FIP: run	●	Profibus communication running
			◐	Communication stopped
			⦿	Profibus configuration missing or faulty
			○	No communication
ERR0 ¹⁾ ERR2 ²⁾	yellow	FIP: error	●	Profibus error
STA0 ¹⁾ STA2 ²⁾	yellow	FIP: send token	◐	Profibus master sends data resp. token
TXD0 ¹⁾ TXD2 ²⁾	yellow	FIP: transmit data	◐	Data are being transmitted
RXD0 ¹⁾ RXD2 ²⁾	yellow	FIP: receive data	◐	Data have been received
ER0 ¹⁾ ER2 ²⁾	yellow	FIP: interface status	●	Profibus faulty (sum error from: short circuit or slave missing or all slaves missing or profibus not connected)
HSK0 ¹⁾ HSK 2 ²⁾	yellow	FIP: handshake	◐	Process data exchange (must be activated in the grid of the parameterized "cycle time for data exchange")
DIA0 ¹⁾ DIA2 ²⁾	yellow	FIP: diagnosis data exchange	◐	Diagnostic data are requested by the parameterized slaves (in error-free operation grid of 500 ms, with slave-failure correspondingly higher)
GC0 ¹⁾ GC2 ²⁾	yellow	FIP: global control	◐	Broadcast message to the slaves to switch their outputs to a predefined state

¹⁾ applies only for the master control element

²⁾ applies only for the processing and communication elements

Operating Elements

Name	Color	Function	Meaning
RES ^{*)}	-	Reset	Reset of the subordinate system elements

^{*)} the Reset button can be activated only if the front panel of the mounting rack is removed

The reset signal of the master control element affects the entire automation unit, this means also the:

- Processing and communication elements
- Protocol elements
- Peripheral elements

The reset signal of a processing and communication element affects its belonging subordinate

- Protocol elements
- Peripheral elements

4.1.2. Peripheral Elements

LEDs on the front panel (refer to [A.3, Peripher](#)) display various states. The arrangement of the LEDs comprises

- System display elements
Display of operating- and error states
- Function display elements
Display of the state of the I/Os

Which states are displayed, is determined by

- the firmware implemented
- the peripheral control module used

System Display Elements

Name	Color	Function	LED	Meaning
RY	yellow	Ready	●	Module ready
ER	red	Error	●	• Sum error (internal error, external error, warning, module failure, failure) • Startup

Function Display Elements

Name	Color	Function	LED	Meaning
D00...D63	yellow	Digital inputs Digital outputs	●	I/O activated
C00...C31	yellow	Digital outputs (command relay)	●	I/O activated
I00... I31	yellow	Digital inputs	●	I/O activated
V00...V23	yellow	Analog inputs Analog outputs	●	I/O activated
GRA	yellow	Group A	●	Command group A activated
GRB	yellow	Group B	●	Command group B activated
OA0	yellow	Output A	●	Output relay 0 activated
OA1	yellow	Output B	●	Output relay 1 activated
CE	yellow	Command error	●	Sum error command output

Operating Elements

Name	Color	Function	Meaning
PBA switch	-	address peripheral element	Setting of the peripheral board address (PBA) parameterized in the TOOLBOX II

4.2. Checks And System Displays

During startup of the AK 1703 ACP system, checks of the hardware and the software are performed by the individual system elements.

If an error is detected, this leads

- to a signaling via the LED-displays
(an overview can be found below)
- depending on the seriousness of the error, possibly to an abortion of the startup

After error-free startup of a system element, further checks are performed.

4.2.1. Master Control Element, Processing And Communication Elements

Startup (step)	RY	ER	WRN	BBD	INT	EXT	COM	HLT	FLR	CPY	Error
Power up or Reset	○	●	○	○	○	○	○	●	●	○	Module defect
Check code memory (flash PROM)	○	●	○	○	○	○	○	●	●	○	PROM error (code) • Module is shutdown • Startup is aborted
Check data memory (RAM) with address error check	○	●	○	○	○	○	○	●	●	○	RAM error (data) • Module is shutdown • Startup is aborted
Check whether flash card present ¹⁾	○	●	○	○	○	○	○	●	●	○	No flash card present
Check of checksums of all parameter and firmware files on the flash card	○	●	○	○	○	○	○	○	○	Ⓜ	Checksum error on the flash card • Diagnostic information is set
Check whether all parameters of the main processor (M-CPU) are present on the flash card ¹⁾	○	●	○	○	○	○	○	●	●	○	Parameter files incomplete • Firmware is shutdown • Startup is aborted

Startup (step)	RY	ER	WRN	BBD	INT	EXT	COM	HLT	FLR	CPY	Error
Check whether all firmware files are present on the flash card ¹⁾	○	●	○	○	○	○	○	○	○	○	Firmware code incomplete
Check, whether firmware code of the lower-level system elements (processing and communication elements ¹⁾ , protocol elements, peripheral elements) is current	○	●	○	○	○	○	○	○	○	○	If not, the firmware code is loaded at a later time
Revision check of the parameters in the parameter memory of the M-CPU	○	●	○	○	○	○	○	○	○	● ¹⁾ ○ ²⁾	If the parameters are not current, they are loaded from the flash card
Check parameter memory (flash PROM)	○	●	○	○	○	○	○	○	○	○	Checksum error of the parameters - Firmware is shutdown - Startup is aborted
Initiate the startup of the lower-level system elements	○	●	○	○	○	○	○	○	○	○	
Check whether the parameterized configuration corresponds with the physical configuration (processing and communication elements ¹⁾ , protocol elements, peripheral elements)	○	●	○	○	○	○	○	○	○	○	Configuration error The incorrectly configured or defective system element is not operational
Loading of the parameters to the lower-level protocol- and peripheral elements	○	●	○	○	○	○	○	○	○	○	Too little memory on the protocol element or peripheral element Firmware on the protocol element or peripheral element is shutdown
Check of the parameter contents	○	●	○	○	○	○	○	○	○	○	Parameter error Diagnostic information is set
Startup concluded without error	●	○	○	○	○	○	○	○	○	○ ³⁾	
Startup concluded with error	●	●	○	○	○	○	○	○	○	○	

¹⁾ applies for the main processor (M-CPU) on the master control element

²⁾ applies for the coprocessor (C0-CPU) on the master control element and on processing and communication elements

³⁾ flashes for approx. 20s after replacement of the master control element resp. of a processing and communication element

Error (startup cancelled)	RY	ER	WRN	BBD	INT	EXT	COM	HLT	FLR	CPY	Note
Module defect	○	●	○	○	○	○	○	●	●	○	Exchange system element
PROM error (code)	○	●	○	○	○	○	○	○	●	○	Exchange system element
RAM error (data)	○	●	○	○	○	○	○	●	○	○	Exchange system element
Parameter files incomplete ^{*)}	○	●	○	○	●	○	○	◐	○	○	Toolbox II diagnostic (initialize automation unit)
Checksum error of the parameters	○	●	○	○	●	○	○	◐	○	○	Toolbox II diagnosis (initialize automation unit)
Parameter error	○	●	○	○	●	○	○	◐	○	○	Toolbox II diagnosis

^{*)} applies for the main processor (M-CPU) of the master control element

Error (startup continued)	RY	ER	WRN	BBD	INT	EXT	COM	HLT	FLR	CPY	Note
No flash card present ¹⁾	●	●	X	X	●	X	X	○	○	○	Toolbox II diagnosis (install flash card, initialize automation unit)
Checksum error on the flash card	●	●	X	X	●	X	X	○	○	○	Toolbox II diagnosis (initialize automation unit or load firmware, acc. to diagnosis)
Firmware code incomplete ¹⁾	●	●	X	X	●	X	X	○	○	○	Toolbox II diagnosis (load firmware automation unit)
Configuration error	●	●	X	●	X	X	X	○	○	○	Toolbox II diagnosis
Too little memory on the protocol element or peripheral element	●	●	X	●	X	X	X	○	○	○	Toolbox II diagnosis
Parameter error	●	●	● ²⁾	X	● ²⁾	X	X	○	○	○	Toolbox II diagnosis (depending on type of error)

¹⁾ applies for the main processor (M-CPU) of the master control element

²⁾ depending on type of error

System element load firmware code	RY	ER	WRN	BBD	INT	EXT	COM	HLT	FLR	CPY	Error
System element is shutdown and the firmware code is loaded from flash card	●	●	●	●	X	X	X	○	○	● ^{*)} ○	
Initiation of the startup of the newly loaded system elements (basic system elements ¹⁾ , protocol elements, peripheral elements)	●	●	●	●	X	X	X	○	○	○	
Startup of the system elements concluded (OK)	●	X	X	X	X	X	X	○	○	○	
Startup of the system elements concluded (error)	●	●	X	X	X	X	X	○	○	○	

^{*)} applies for the main processor (M-CPU) of the master control element

4.2.2. Protocol Elements

Startup (step)	ER	Error
Reset, power up or startup initiated by the higher-level system element	●*)	Module defect
Check code memory (flash PROM)	●*)	PROM error (code) - Module is shutdown - Startup is aborted
Check data memory (RAM)	●*)	RAM error (data) - Module is shutdown - Startup is aborted
Reception of the parameters from the higher-level system element	●*)	Too little memory - Firmware is shutdown - Startup is aborted
Check of the parameter contents	●*)	Parameter error Diagnosis information is set
Startup concluded (OK)	○	
Startup concluded (error)	●*)	

*) in addition the system-LED "ER" lights up on the basic system element

Action after startup	ER	Error
Setup of the communication connection	●*)	Communication failure Communication fault
Communication established	○	

*) in addition the system-LED "ER" lights up on the basic system element

Error (startup cancelled)	ER	Note
Module defect	●*)	Exchange system element
PROM error (code)	●*)	Exchange system element
RAM error (data)	●*)	Exchange system element
Too little memory	●*)	Toolbox II diagnosis
Parameter error	●*)	Toolbox II diagnosis

*) in addition the system-LED "ER" lights up on the basic system element

Error (startup continued)	ER	Note
Parameter error	●*)	Toolbox II diagnosis
Communication failure	●*)	Toolbox II diagnosis
Communication fault	●*)	Toolbox II diagnosis

*) in addition the system-LED "ER" lights up on the basic system element

4.2.3. Peripheral Elements

Startup (step)	RY	ER	Error
Reset, power up or startup initiated by the higher-level system element	○	●*)	Module defect
Check code memory (flash PROM)	○	●*)	PROM error (code) - Module is shutdown - Startup is aborted
Check data memory (RAM)	○	●*)	RAM error (data) - Module is shutdown - Startup is aborted
Reception of the parameters from the higher-level system element	○	○*)	Too little memory - Firmware is shutdown - Startup is aborted
Check of the parameter contents	○	○*)	Parameter error Diagnostic information is set
Startup concluded (OK)	●	○	
Startup concluded (error)	●	●	

*) in addition the system-LED "ER" lights up on the basic system element

Error (startup cancelled)	RY	ER	Note
Module defect	○	●*)	Exchange system element
PROM error (code)	○	●*)	Exchange system element
RAM error (data)	○	●*)	Exchange system element
Too little memory	○	○*)	Toolbox II diagnosis
Parameter error	○	●*)	Toolbox II diagnosis

*) in addition the system-LED "ER" lights up on the basic system element

Error (startup continued)	RY	ER	Note
Parameter error	●	●*)	Toolbox II diagnosis

*) in addition the system-LED "ER" lights up on the basic system element

4.3. Diagnosis

4.3.1. General

The following listing explains the different error categories and how the error messages are logged.

- Hardware errors

Monitoring of the individual system elements, communication and time synchronization by the firmware of the target system in operation. Error information takes place by

- Logging via diagnostic function (online)
- Activation of the red error-LED on the master control element

A list of error messages which are indicated in the diagnostic can be found in the manual of the respective master control element (example: *System Element Manual CP-2010/CPC25, Appendix "Diagnostic Information"*).

Information about the diagnostic classes is included in the *TOOLBOX II User Manual, chapter "Diagnostic"*.

- Software errors

- Parameter setting

The input of the parameters with the help of the TOOLBOX II tools is subjected to - as far as logically recordable - parameter-specific conventions or plausibility checks. Faulty inputs are notified or directly prevented with the entry.

If a - in principle permitted - parameter setting, that however does not match the physically present system elements, is loaded into the target system, the method of procedure is as with hardware errors.

- Programming

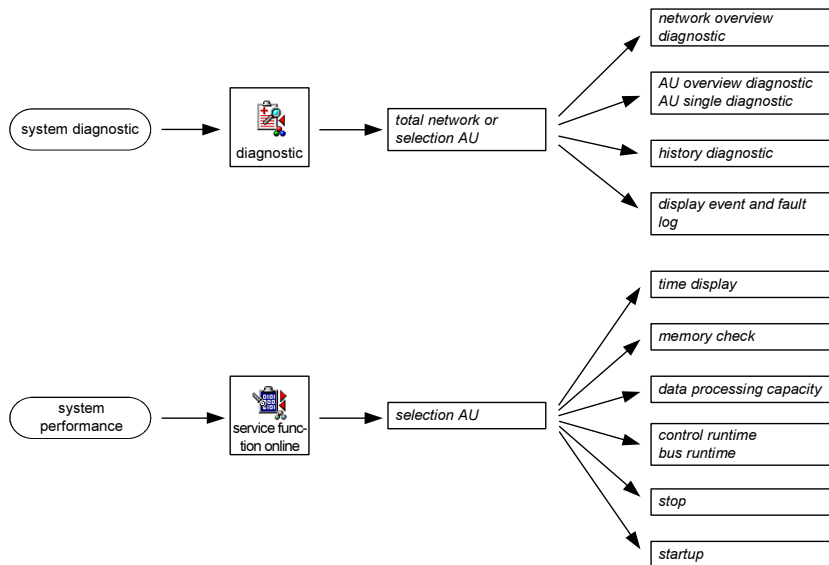
Monitoring by the CAEx *plus* compiler and logging in an error information list during compiling of the application program code.

- Operator errors

Are largely excluded due to plausibility checks or queries by the TOOLBOX II. Naturally, all operator inputs, such as for example the loading of parameters or the online testing of application programs, are subject to the responsibility of the user. Profound knowledge of the automation technique in general, as well as the documents mentioned in section [Literature](#), is presumed.

4.3.2. System Diagnosis

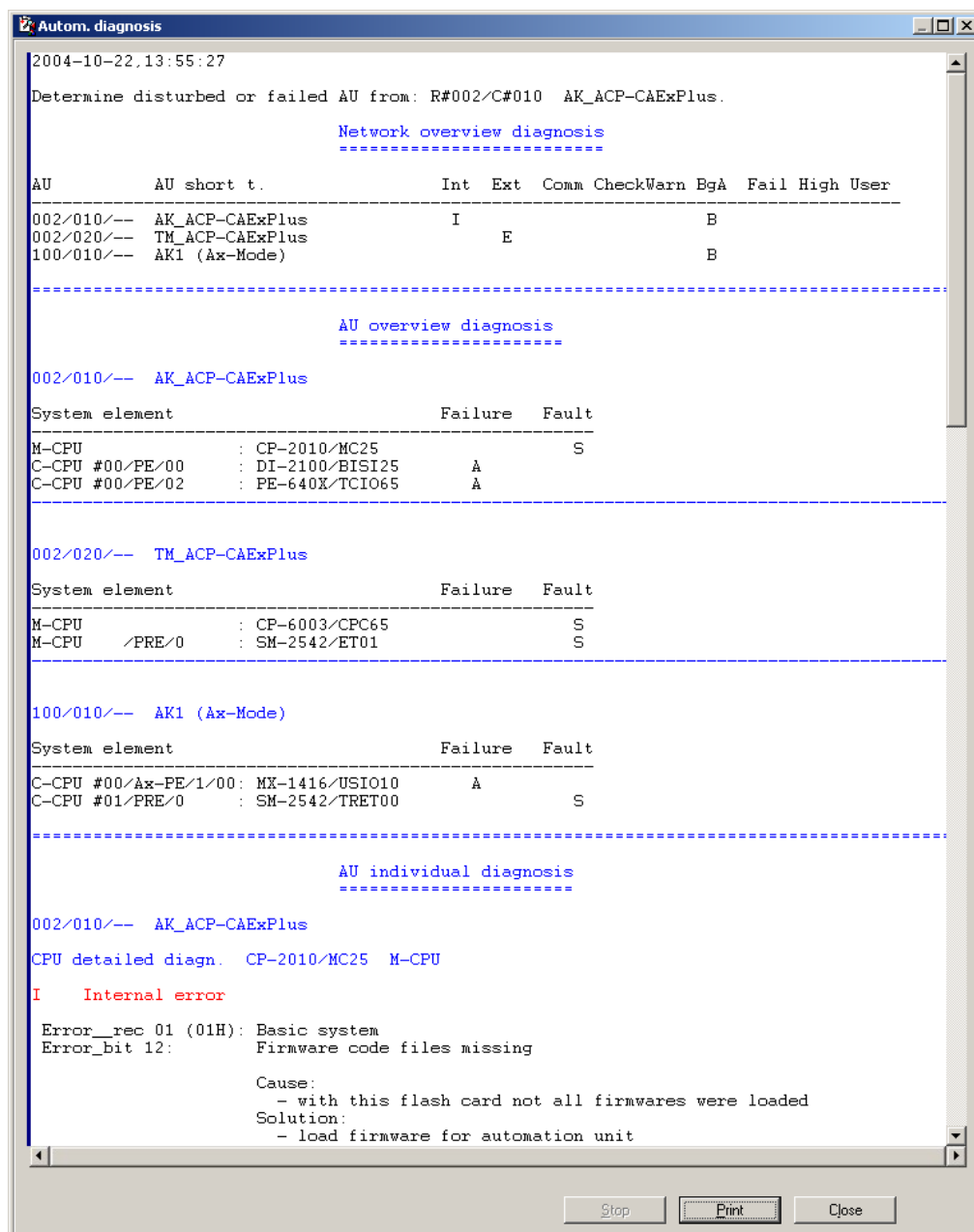
AK 1703 ACP contains extensive diagnostic functions for monitoring the system. Since the firmware automatically executes the appropriate error monitoring routines, for this no settings of any kind by the user are necessary.



With the tool "Diagnostic", system states and errors can be read out and displayed in plain text and printed out if required (online).

The following functions are available:

- **Acknowledge network**
With the function ACKNOWLEDGE NETWORK, the signals of all automation units located in the network are reset, insofar as they are no longer present.
- **Network overview diagnostic**
The error messages of all automation units in the network are output in a report window with the function NETWORK OVERVIEW DIAGNOSTIC (sum information).
- **AU overview diagnostic**
The error messages of all automation units in the network are output in a report window with the function AU OVERVIEW DIAGNOSTIC (sum information for each system element able to report).
- **History**
The history of error messages is output in a report window with the function HISTORY DIAGNOSTIC.
- **Automatic diagnostic**
An overview of the error messages of all automation units in a network as well as their system elements is output in a report window with the function AUTOMATIC DIAGNOSTIC. The outputs are structured hierarchically in
 - Network overview diagnostic
 - AU overview diagnostic
 - AU single diagnostic
 - History (if preset)



Further details about the diagnostic functions can be found in the *TOOLBOX II User Manual*, chapter "Diagnostic".

The technical description of the diagnostic functions can be found in the manual *ACP 1703 Common Functions System and Basic System Elements*, chapter "System Services", section "Diagnostic".

4.3.3. System Performance

Information about the system performance of a selected automation unit can be interrogated (online) with the tool "Service Function Online":

- Display equipped system elements
- Read serial numbers
- Display/delete data flow routing
- Shut down selected system elements
- Shut down selected automation unit
- Startup selected system elements
- Startup automation unit
- Display time
- Memory check
- Data processing capacity
- Control- and bus runtime

Further details about the service functions can be found in the *TOOLBOX II User Manual*, chapter "Service Function Online".

4.3.4. Diagnosis Of The Open-/Closed-Loop Control Function

With the tool "CAEx plus" the following status information can be displayed for each task with the function ONLINE-TEST:

- Parameterized cycle time
- Current runtime
- Minimum/maximum runtime
- Number of runtime time-outs
- Bus runtime

Further Information can be found in the manual *ACP 1703 Common Functions System and Basic System Elements*, chapter "Automation", section "Open-/Closed-loop Control Function", section "Display Status Information".

Errors that can occur with the creation of the function diagram (for example the linking of a binary signal to an analog input of a function) are displayed in a separate report window ("POU error check"):

- Automatically with the incorrect input
- Manually with the function EARLY ERROR DETECTION (Pop-up menu in the function diagram). All editing errors can be displayed at any time.

The exact details can be found in the *CAEx plus User Manual*, chapter "Editors", section "FBD Editor", section "Operation in FBD Editor".

Errors On Engineering

Errors that can occur on the function diagram engineering are displayed centrally in the "Error-state viewer" of "CAEx *plus*". It essentially concerns thereby

- Installation binary information
- Export/Import binary information
- Project management binary information
- FBD-editor binary information
- Code generation binary information
- Object-related binary information
- Binary information of the object assistants
- Binary information of the typical import
- Signal list binary information
- Binary information from hardware signal databases

The binary information are generated after the tripping of a function (for example "Code generation").

The exact details can be found in the *CAEx plus User Manual*, chapter "Additional Products", section "Error-State Viewer".

Errors With The Target System Connection

Errors that can occur with the target system connection are displayed centrally in the "Error-state viewer" of "CAEx *plus*". It concerns thereby

- General messages of the target system connection
- Messages of the code generator
- Messages of the control panel
- Messages of the ONLINE applications
- Messages of the "Open Operating System"
- Messages of the target-system integration into the project management
- Messages from "openPLC"

The exact details can be found in the *CAEx plus User Manual*, chapter "Target-System Connection", section "Program Messages for Target-System Connection".

4.4. Maintenance Of The Hardware

4.4.1. Guideline For The Replacement Of Modules

By means of the modular design of AK 1703 ACP the replacement of a faulty module is achievable in an easy manner:

- The peripheral cables can be detached without tools.
- The modules are detached from the rack with the unlock tool TA2-105 (scope of delivery).
- Serial interface modules and connection boards can be detached without tool.
- The user data (parameters and application program) are stored non-volatile on the flash card.
- The flash card can be withdrawn from the removed master control module.
- For the commissioning of a new module the TOOLBOX II is not required.



Note

For the replacement of system elements a shutoff of the power supply is not required. By means of that, the function of the subsystems which are not affected by the replacement is maintained.



Warning

For peripheral elements that operate with voltages > 60 V (e.g. DI-2111) care must be taken that manipulation on the peripheral connectors may only be carried out in a de-energized state.

The detailed hints for the module handling can be found in the manual *AK 1703 ACP Installation*, chapter "Installation and Removal of Modules" and chapter "Setup of external Communication Connections".

More exact technical information about the module replacement can be found in the manual *ACP 1703 Common Functions System and Basic System Elements*, chapter "System Services", section "Data Storage on Flash Card".

4.4.2. Recognize And Replace A Faulty Module

During operation AK 1703 ACP monitors the configured hardware. A faulty or failed system element is signaled as follows:

- Red ER-LED on the master control element
- Function DIAGNOSTIC | AUTOMATIC DIAGNOSTIC
- Function DIAGNOSTIC | AU OVERVIEW DIAGNOSTIC

If a hardware error is present, a replacement of the respective system element by the same type (with the same firmware, even if a later revision level is present) is necessary.

If the relevant type is no more deliverable, then the appropriate successor must be used (see [1.2.2, Replaced Modules](#)).

When replacing a system element with another type, a change of the project data with the TOOLBOX II is required (the necessary functionality of the system must be recovered by means of re-parameterization). After the loading of the changed configuration into the target system, a startup of the system must be performed.

4.4.2.1. Replacement Of The Master Control Element

After removing the master control element from the mounting rack, the function of the system is naturally not present. An exception forms an "autarchic" subsystem, whose function independently of the master control element is running in a processing and communication element.

The flash card can be taken out of the removed master control element, and being plugged into the new master control element.

After plug-in of the new master control element the startup takes place. Thereby, the project data (parameters and application program) are being loaded from the flash card.

4.4.2.2. Replacement Of A Processing And Communication Element

After removing the processing and communication element from the mounting rack, the function of the affected subsystem is naturally not present.

After plug-in of the new processing and communication element, this is detected by the master control element, and the project data (parameters and application program) are being loaded from the flash card of the master control element.

Since for the replacement of a processing and communication element a shutoff of the power supply is not required, the function of other basic system elements and their subordinate system elements is maintained.

4.4.2.3. Replacement Of A Serial Interface Module

For the replacement of a serial interface module first the basic system element (carrier module) must be removed from the mounting rack. For this, please consider the previously mentioned consequences for the system.

Upon replacement of a serial interface module the parameters already stored on the flash card of the master control element are automatically utilized on the new module.

4.4.2.4. Replacement Of A Peripheral Element

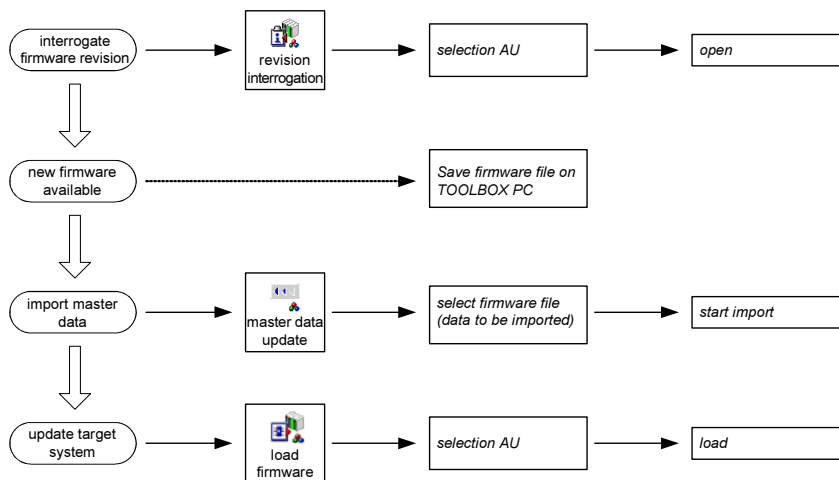
After the replacement of a peripheral element the parameters already stored on the flash card of the master control element are automatically utilized on the new module.

4.5. Update Of The System

In AK 1703 ACP, new system elements or new revisions of system elements can be integrated, if required. The updating of the system takes place in steps:

1. Import master data into the TOOLBOX II
 - Firmwares for the tools of the TOOLBOX II
 - Firmware codes for loading into the target system
2. Load firmware codes into the target system
 - Processing of the updated system elements or functions by executing the firmware codes

Procedure For The Updating Of System Elements



4.5.1. Interrogate Firmware Revision

The firmware revisions of all configured system elements (basic system element, peripheral element, protocol elements) can be displayed on the screen and printed out for a selected automation unit with the tool "Revision Interrogation".

Further details about the tool can be found in the *TOOLBOX II User Manual, chapter "Revision Interrogation"*.

4.5.2. Import Master Data Into The TOOLBOX II

The importing of the master data into the TOOLBOX II is performed with the tool "Master Data Update". For this, corresponding software revisions are issued by SIEMENS as files.

The details about the operation can be found in the *TOOLBOX II User Manual, chapter "Master Data Update"*.

4.5.3. Update Target System

The TOOLBOX II contains all current firmware revisions after its installation (based on the issue date of the installed version).

Firmware revisions, that have been imported into the TOOLBOX II later with the tool "Master Data Update", can be loaded into the connected target systems with the tool "Load Firmware".

The loading of the firmware refers to all loadable system elements of the automation unit:

- Master control element
- Processing and communication elements
- Peripheral elements
- Protocol elements

For the execution, the configuration possibilities listed under section [1.4.3.1, Transfer Of Engineering Data To AK 1703 ACP](#) are applicable. When you have connected the PC with the master control element, you can begin with the loading of the firmware.

The details thereto can be found in the *TOOLBOX II User Manual, chapter "Load Firmware"*.

4.5.3.1. Errors When Loading The Firmware

When loading the firmware into flash PROM this is written completely new. For a perfect functioning of the master control element, it is therefore absolutely necessary that this procedure completes without errors.

If the tool "Load Firmware" terminates with an error indication, start the loading procedure again and attempt to perform it without error. If an error occurs again, the master control element is no longer capable of functioning. Check the LED display.

The following situations cause the master control element to be no longer capable of functioning:

- Voltage failure while loading procedure is running
- Abortion by the user while loading procedure is running
- Failure (defect, shutdown) of the PC
- Abortion due to hardware error
- Abortion or excessively long fault of the communication connection between PC and AK 1703 ACP

Further technical information can be found in the manual *ACP 1703 Common Functions System and Basic System Elements, chapter "System Services", sections "Loading Operation" and "Firmware Shutdown"*.



Note

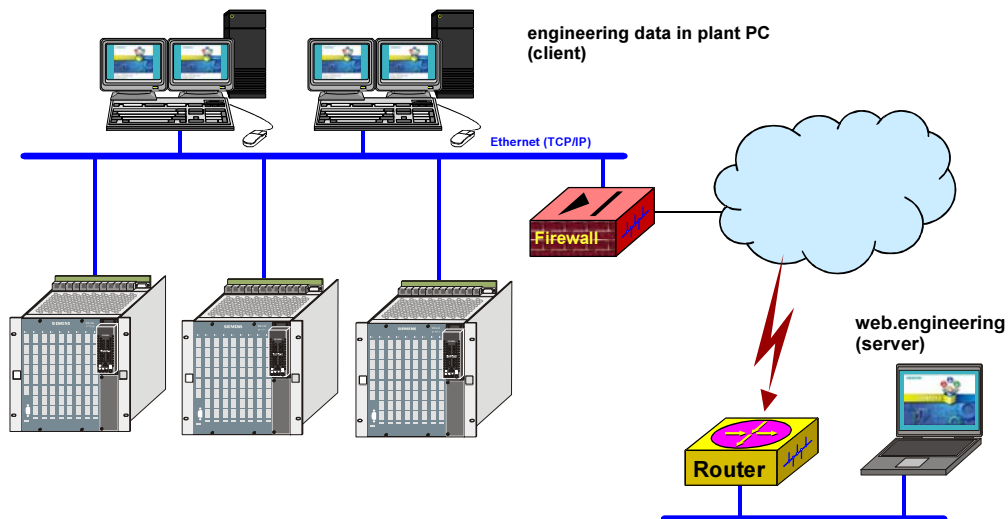
A master control element that is no longer capable of functioning can be recognized by the fact, that after a restart of the AK 1703 ACP the yellow RDY LED does not light up and the red ER LED and the yellow HLT LED do not turn off any more. An access with the TOOLBOX II is no longer possible in this case.

In this case please consult:

SIEMENS AG
 Energy Customer Support Center
 Phone: + 49 180 524 7000
 Fax: + 49 180 524 2471
<mailto:support.energy@siemens.com>
<http://www.siemens.com/energy-support/en>

4.6. Remote Maintenance

With the help of the "web.engineering" option, one can remotely operate (remote maintenance session) a remote TOOLBOX II that can be reached via a data connection (Modem, ISDN, LAN/WAN) with a controlling PC (remote maintenance center).



At "web.engineering" all TOOLBOX II clients work with a central web server which is running the TOOLBOX II. Thereby the TOOLBOX II is operated in the Internet Explorer (as of Version 5.0).

As server, a Windows server with Terminal Server and Web Service must be installed. On this server a normal TOOLBOX II installation is carried out afterwards. On the client no installation is required for this.

A detailed description of the configuration of the server and the client can be found in the *TOOLBOX II User Manual*, chapter "web.engineering".

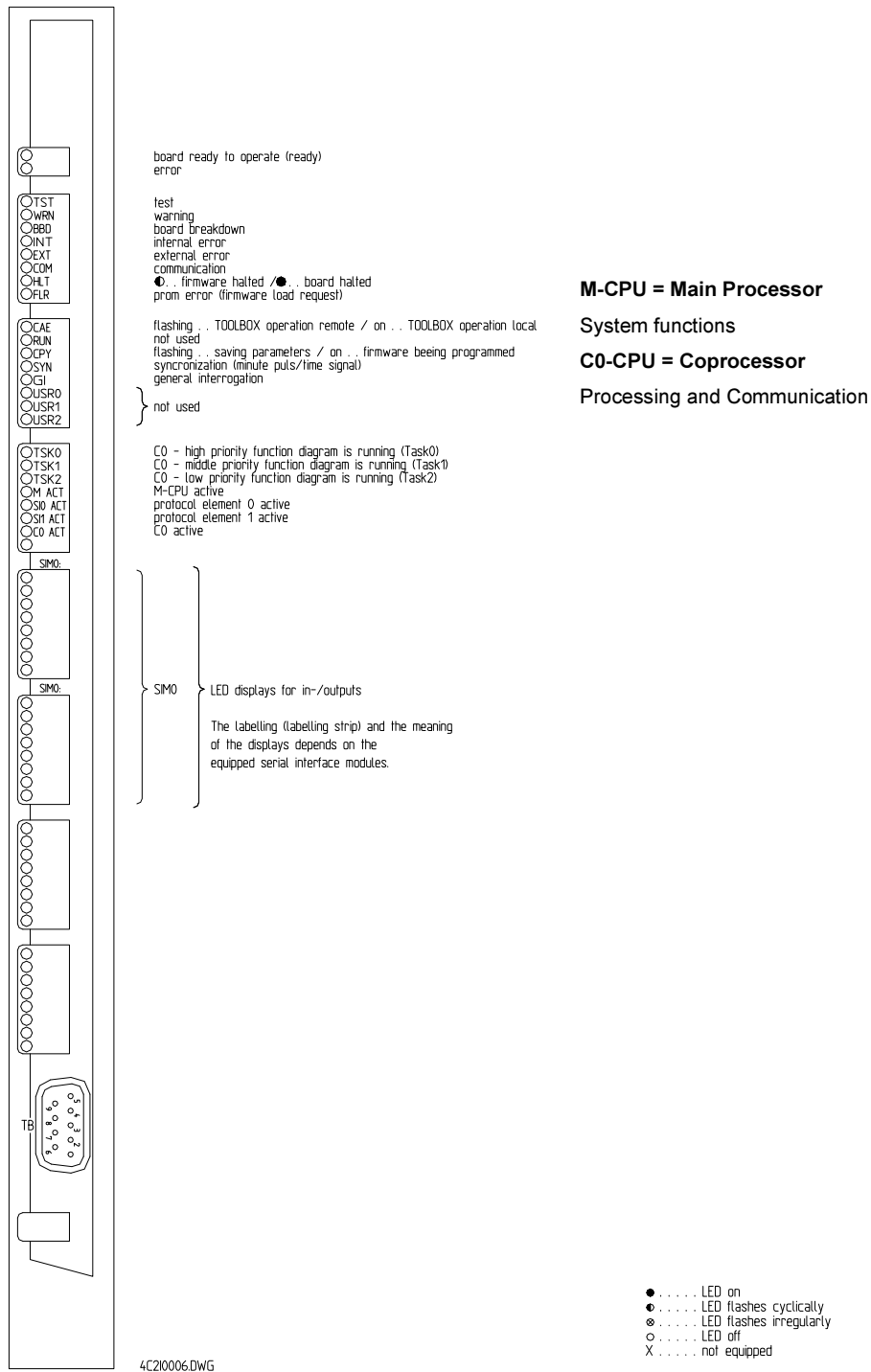
A. Front View of the Modules

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A.4.	Power Supply	81
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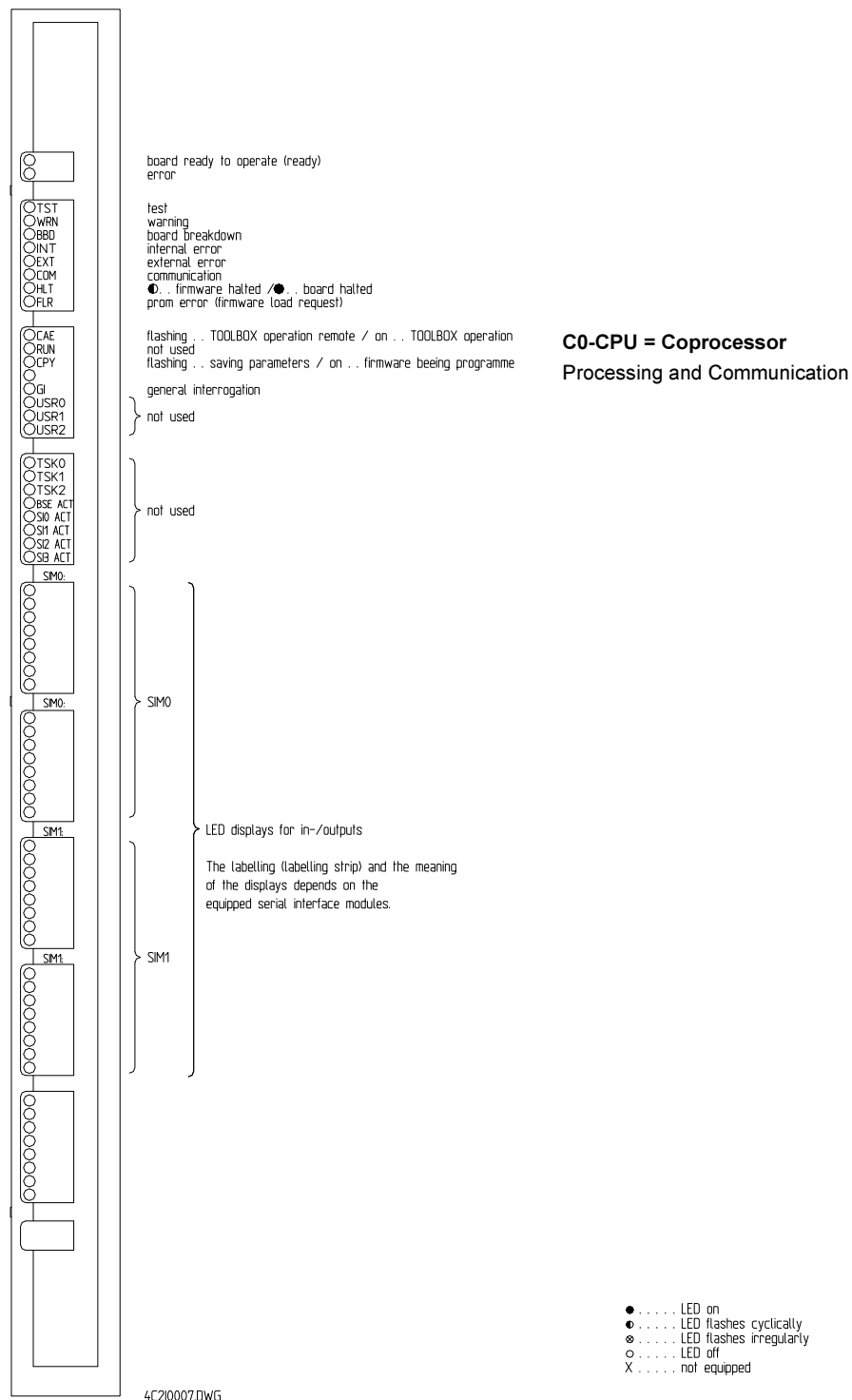
A.1. Master Control Unit

A.1.1. CP2010/CPC25



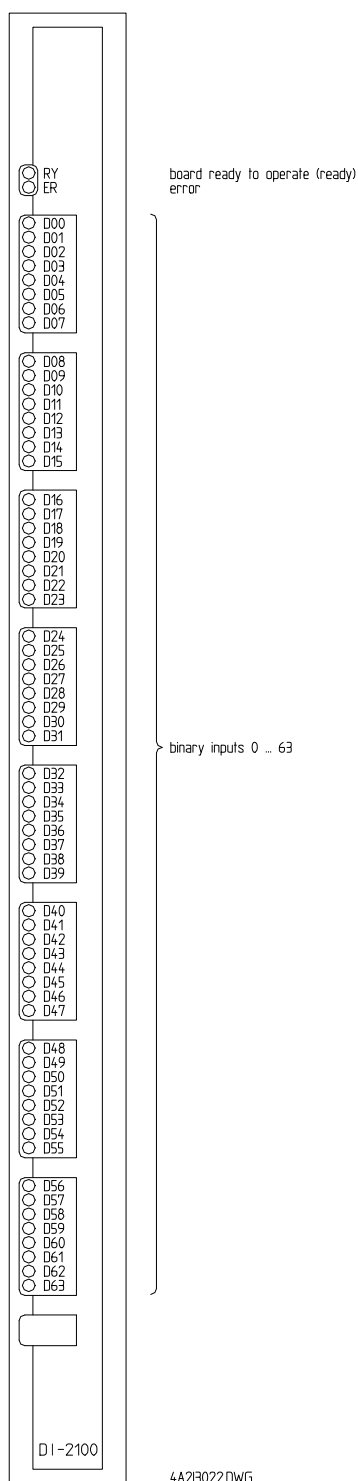
A.2. Processing And Communication

A.2.1. CP-2017/PCCX25

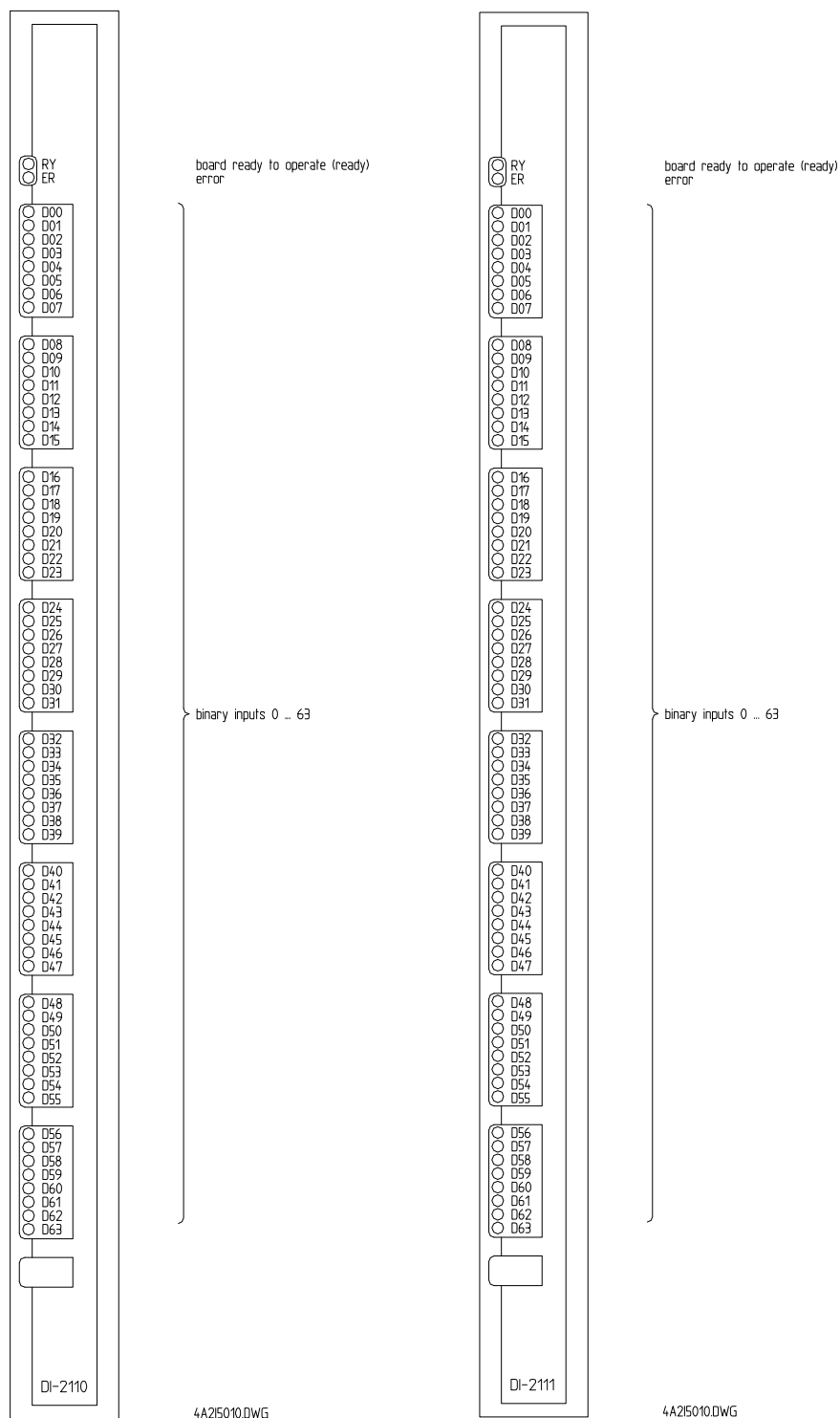


A.3. Periphery

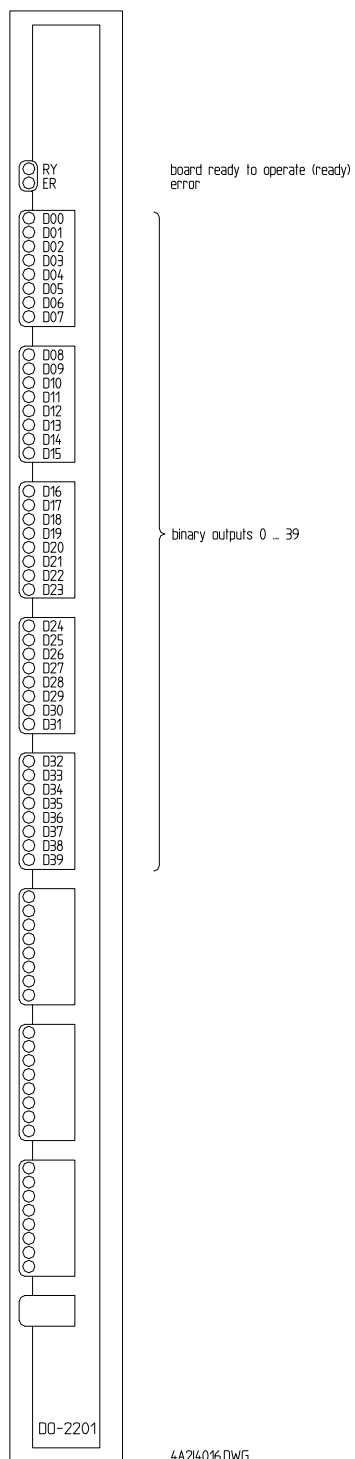
A.3.1. DI-2100/BISI25



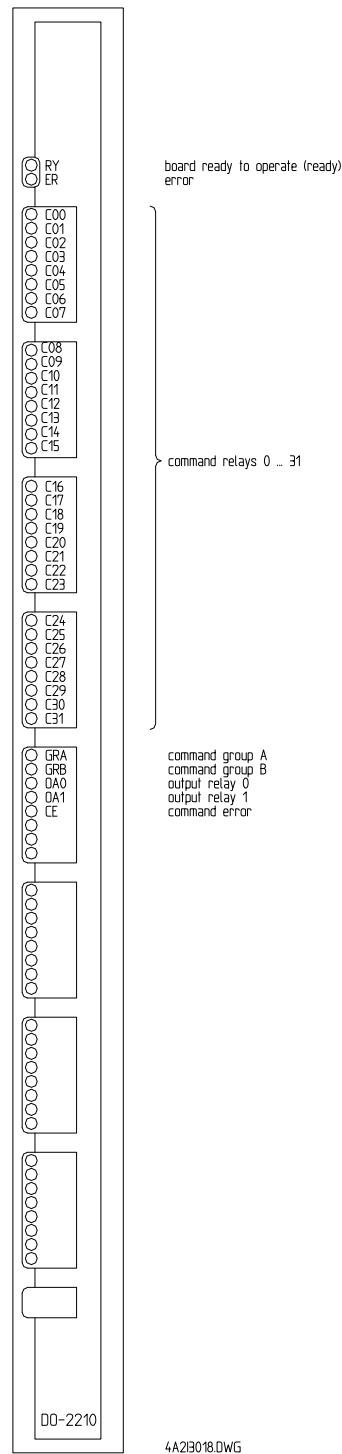
A.3.2. DI-2110/BISI26, DI-2111/BISI26



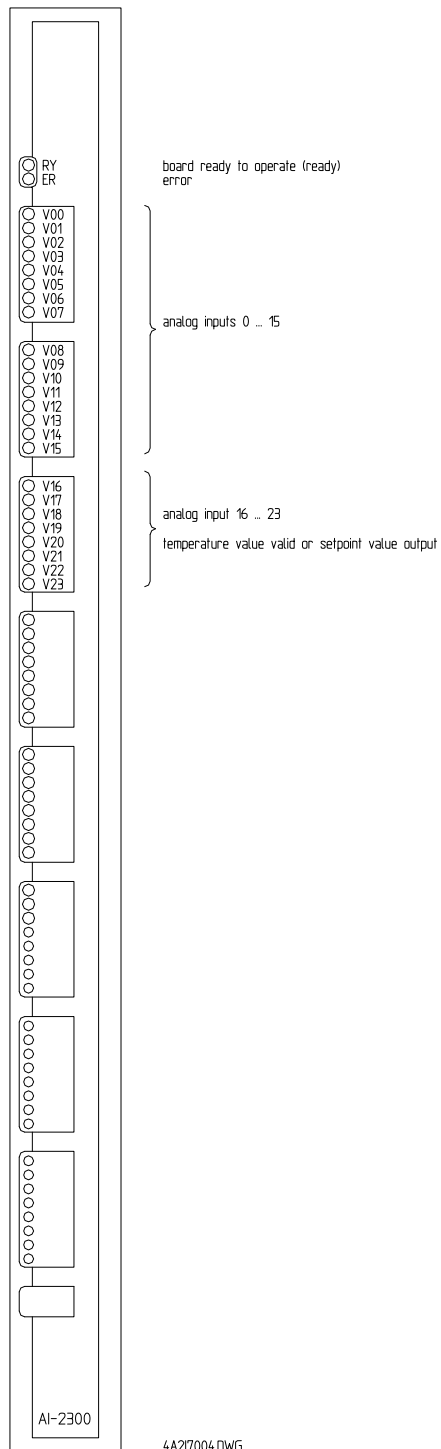
A.3.3. DO-2201/BISO25



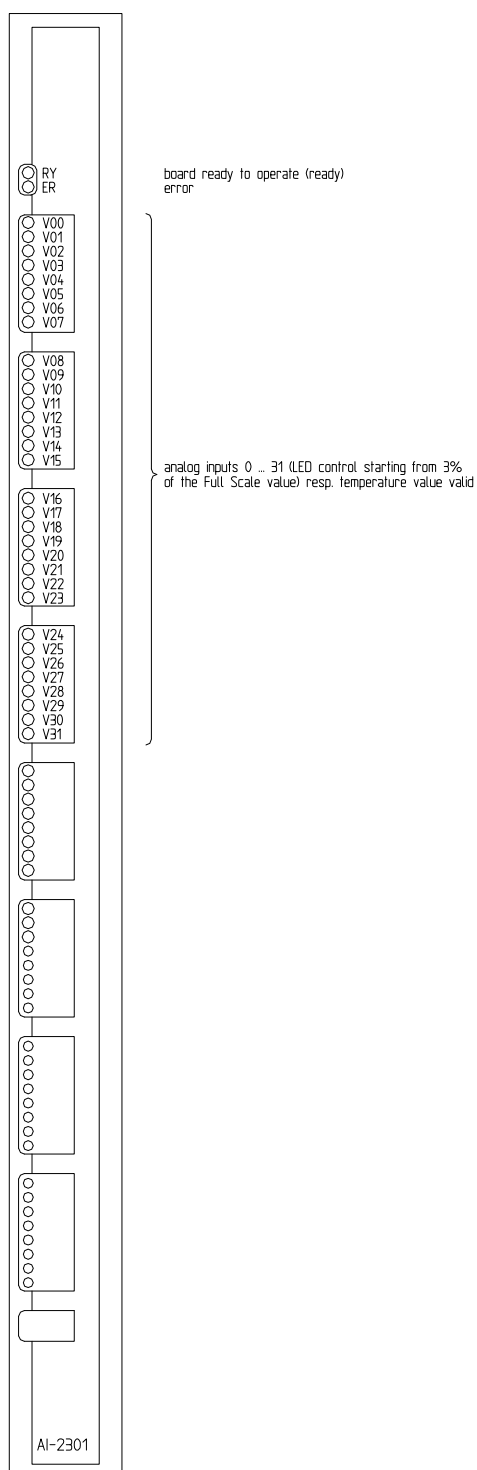
A.3.4. DO-2210/PCCO2x



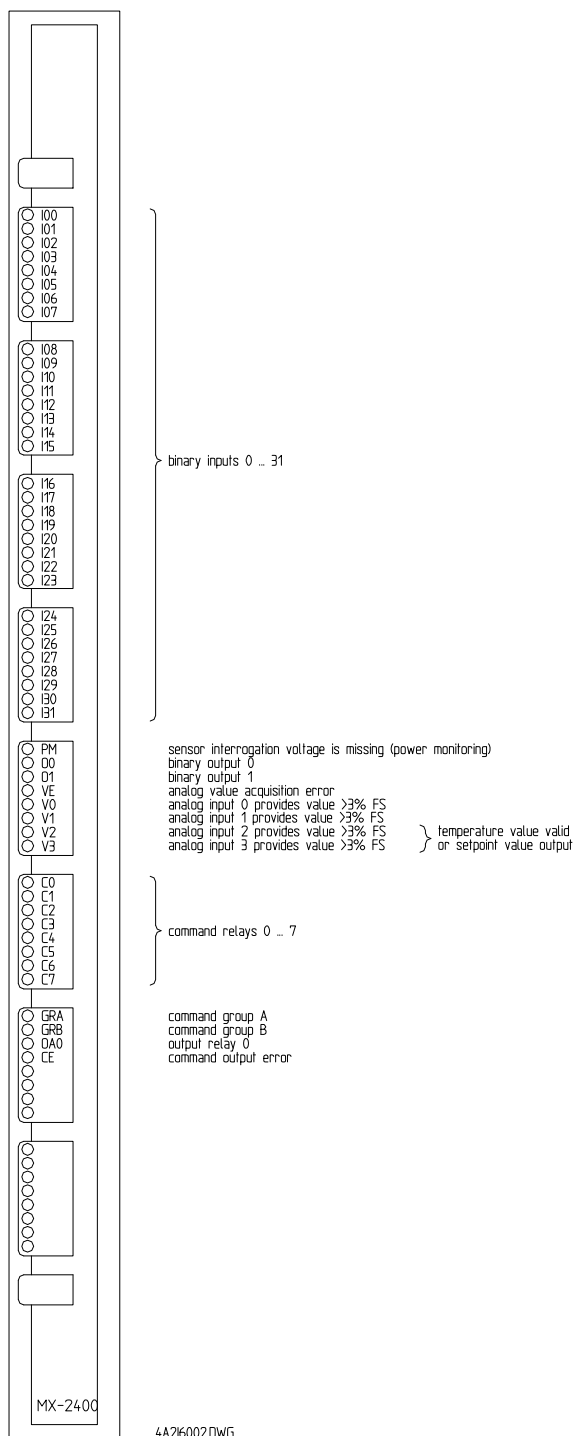
A.3.5. AI-2300/PAS125



A.3.6. AI-2301/TEMP25

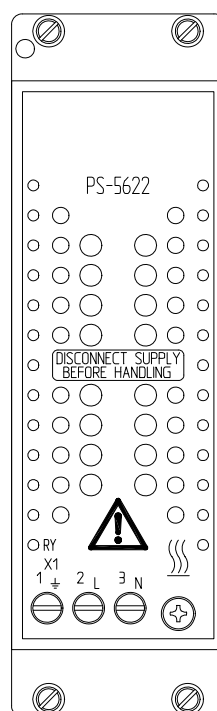
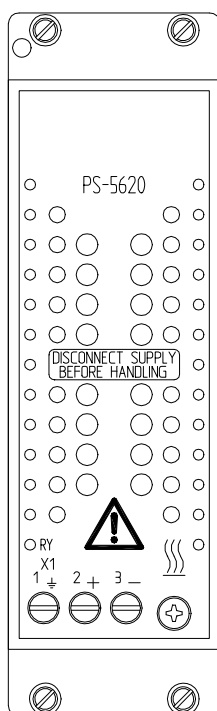


A.3.7. MX-2400/USIO2x



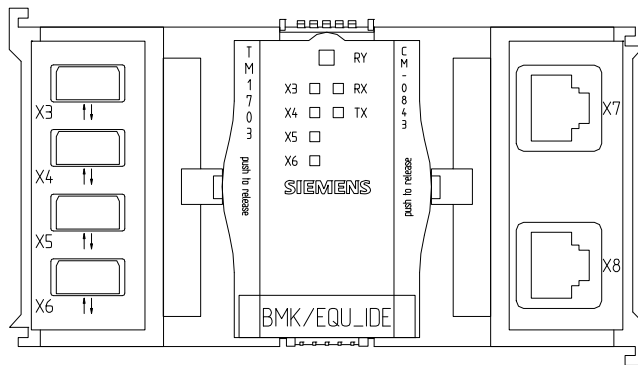
A.4. Power Supply

A.4.1. PS-5620, PS-5622



A.5. Bus Interface

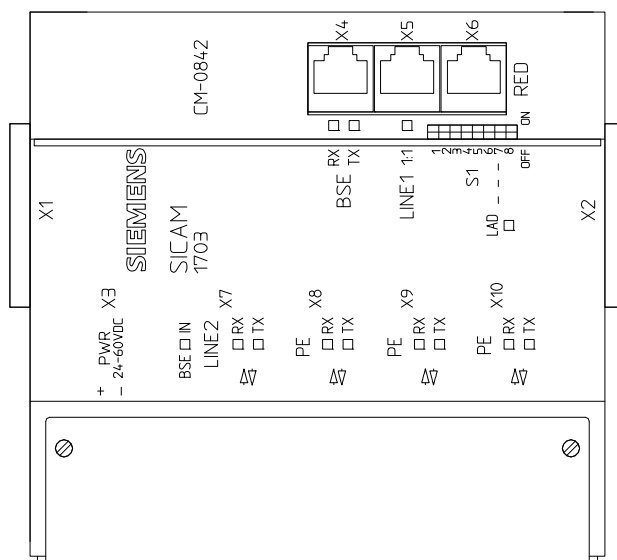
A.5.1. CM-0843



RY supply voltage OK
 RX receive data from master
 TX transmit data to master
 X3 power supply over interface X3 OK
 X4 power supply over interface X4 OK
 X5 power supply over interface X5 OK
 X6 power supply over interface X6 OK

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A.5.2. CM-0842



Literature

AK 1703 ACP Installation	DC2-009-2
AK 1703 ACP Functional Description	MC2-011-2
CM-2832 Data Sheet	MC2-005-2
CM-2835 Data Sheet	MC2-007-2
CP-2010/CPC25 System Element Manual	DC2-005-2
CP-2017/PCCX25 System Element Manual	DC2-015-2
DI-2100/BISI25 System Element Manual	DC0-001-2
DI-211x/BISI26 System Element Manual	DC0-005-2
DO-2201/BISO25 System Element Manual	DC0-017-2
DO-221x/PCCO2x System Element Manual	DC0-003-2
AI-2300/PASI25 System Element Manual	DC0-007-2
AI-2301/TEMP25 System Element Manual	DC0-051-2
MX-2400/USIO2x System Element Manual	DC0-019-2
ACP 1703 Common Functions System and Basic System Elements	DC0-015-2
SICAM 1703 Common Functions Peripheral Elements according to IEC 60870-5-101/104	DC0-011-2
SICAM 1703 IEC 60870-5-101/104 Interoperability	DC0-013-2
TOOLBOX II User Manual	D30-001-2
TOOLBOX II CAEx <i>plus</i> User Manual	D30-008-2
ACP 1703 Platforms Configuration Automation Units and Automation Networks	DC0-021-2
PC Products Preference List	D95-003-1
PE-641x/TCIO66 System Element Manual	DC6-037-2
PE-641x/USIO66 System Element Manual	DC6-033-2
DI-111_/BISI15 Data Sheet	MA1-121-2
AI-1304/TIPP16 Data Sheet	MA1-108-2
MX-141_/USIO15 Data Sheet	MA1-123-2

